



# OcuTrap Knowledge Base

Complete user guide for the OcuTrap R1 smart wildlife trap.

Online docs: [docs.ocutrap.com](https://docs.ocutrap.com)

Support: [support@ocutrap.com](mailto:support@ocutrap.com)

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# OcuTrap Knowledge Base

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|                    |                                                                 |
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| Getting Started    | <a href="#">introduction.md</a>                                 |
| Account            | <a href="#">resetting-password.md</a>                           |
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## Website Status

<https://ocutrap.statuspage.io/>

# Getting Started

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## Introduction



### What is OcuTrap?

- **Innovative OcuTrap Technology:** Transforming the approach to trap monitoring, management, and control.
- **Significant Time and Cost Savings:** Drastically reduces the need for frequent trap checks, saving both time and resources.
- **Instant Mobile Alerts:** Get notified immediately upon the capture of your target animal, enabling efficient trap management.
- **Smart Control Features:** Facilitates the precise capturing of target animals through advanced technology.
- **Remote Operation Capability:** Manage trap doors from anywhere in the world, significantly reducing the risk of injuries from accidental encounters with the trapped animals.

First time using you OcuTrap? Check out

[safety-information.md](#)

## Set-up Tutorial

There are two main parts to getting started.

|                                      |  |
|--------------------------------------|--|
| <a href="#"><u>setting-up.md</u></a> |  |
| <a href="#"><u>app</u></a>           |  |

## Using OcuTrap

- Accessing web-portal ([base.ocutrap.com](http://base.ocutrap.com))
- Exploring different settings

## Hardware Set Up

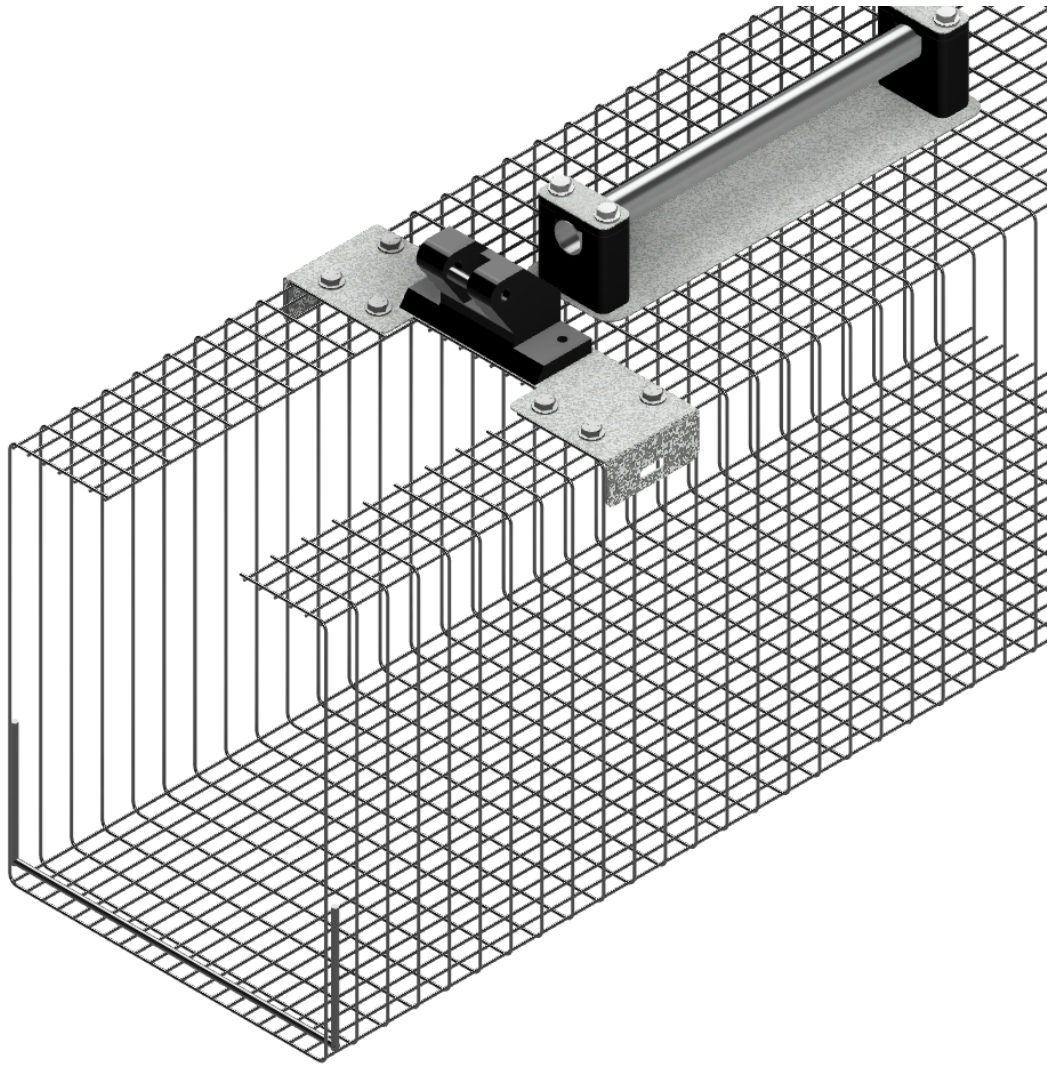
### OcuTrap R1 Video Set Up

► **This page has a video.** Watch it online at [docs.ocutrap.com](http://docs.ocutrap.com).

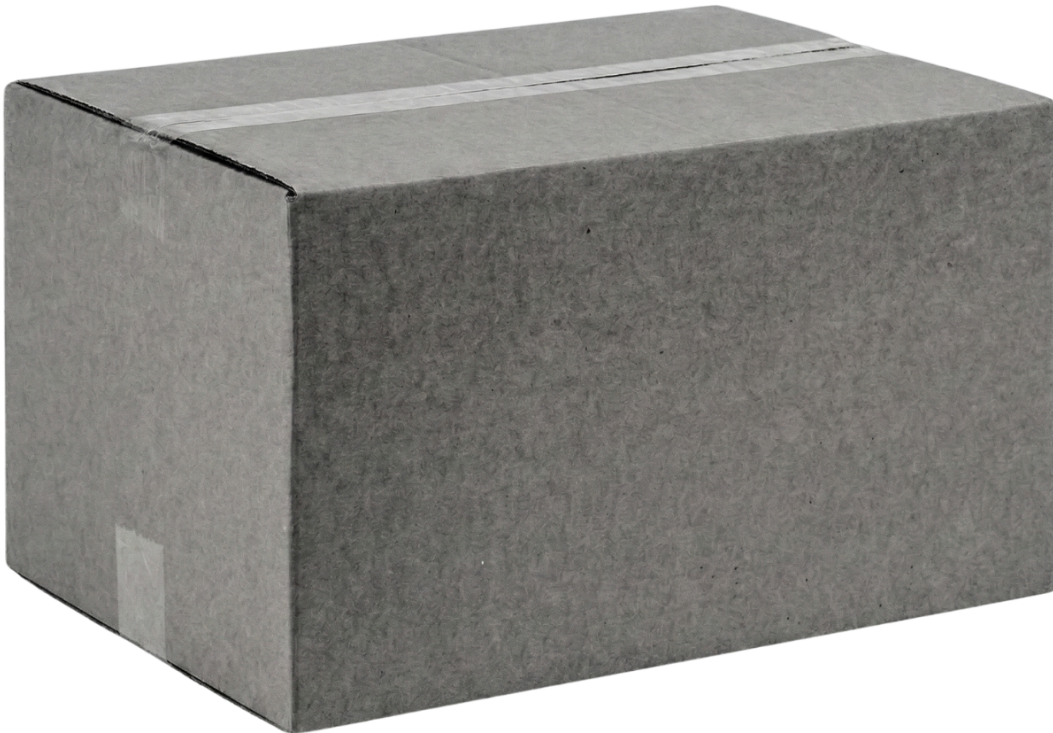
## Unboxing and Initial Inspection

Carefully unpack your OcuTrap R1 and check that all components are included:

### Cage



**Parts Box**



Examine each item for any visible damage. If anything is missing or appears damaged, halt installation and contact OcuTrap Support with your Trap ID.

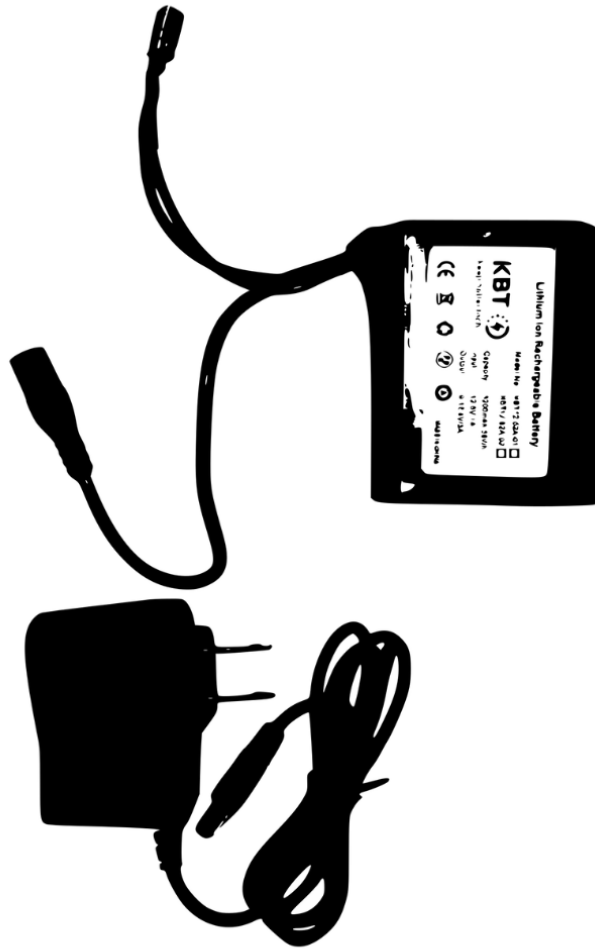
### Hardware Setup

#### Unpack the Components

1. Remove the OcuTrap Device from its packaging.
2. If anything appears damaged, contact [support@ocutrap.com](mailto:support@ocutrap.com) for assistance.

#### Charge the Battery

1. Fully charge the blue battery until the light is green using the accompanying charger found in the small white box.



Introduction Welcome to the OcuTrap Setup Manual. This guide will walk you through the process of assembling your OcuTrap step by step. The setup process is divided into four main sections: Door, Handle, POD Setup, and App Setup

Carefully follow each step to ensure your OcuTrap functions correctly and efficiently



## Section 1: Door Setup

### ***Step 1: Components Needed for Setup***

Before beginning, ensure you have the following parts for the door assembly:

#### **Door**

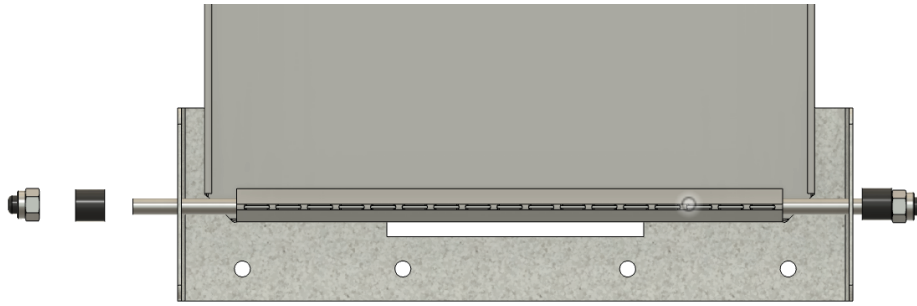
- 2x Brackets (top locking mechanism)
- 2x Black spacers
- 2x Black capped nuts
- 1x Metal door
- 1x 12" Rod
- 1x Nut Driver
- 1x Nut Assembly Tool (figure right)

#### **Motor**

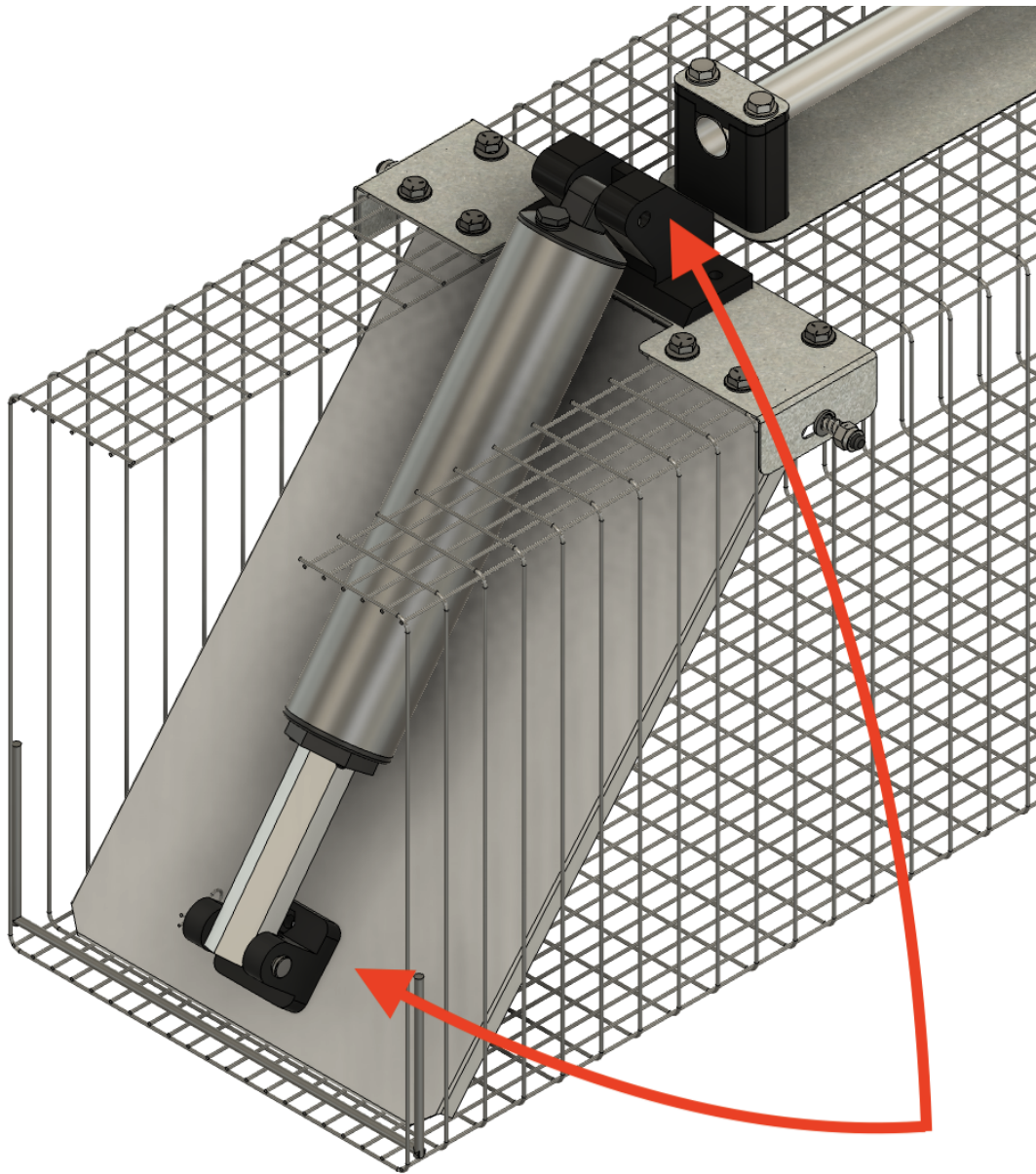
- 1x Motor
- 2x Pins
- 2x Clevises

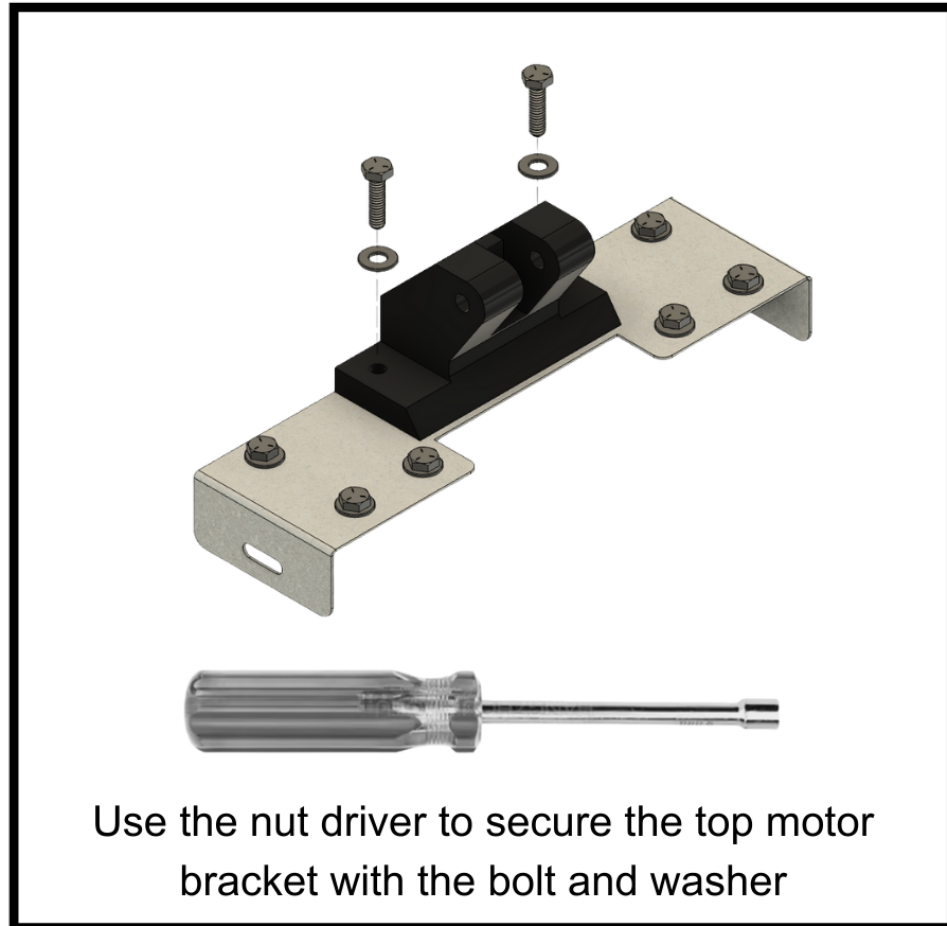
### ***Step 2: Setup the Door Mechanism***

1. Align the metal door inside the trap.
2. Thread the metal rod through the oval slot in the metal bracket attached to the top of the solid metal trap door.
3. On each end of the rod:
  1. Place a black spacer.
  2. Secure it with the black capped nut on both sides
  3. Use the nut assembly tool and the nut driver on each end to tighten the nut until snug.

***Step 3: Setup the Motor***

1. Install top bracket with washers and bolts. Tighten with nut driver.
2. Use the pins and clevises to secure the motor to the door at both the top and bottom attachment points.
3. Feed the cable through the metal handle.
4. Verify that all components are securely fastened.
5. Check that the door moves smoothly and is properly aligned.



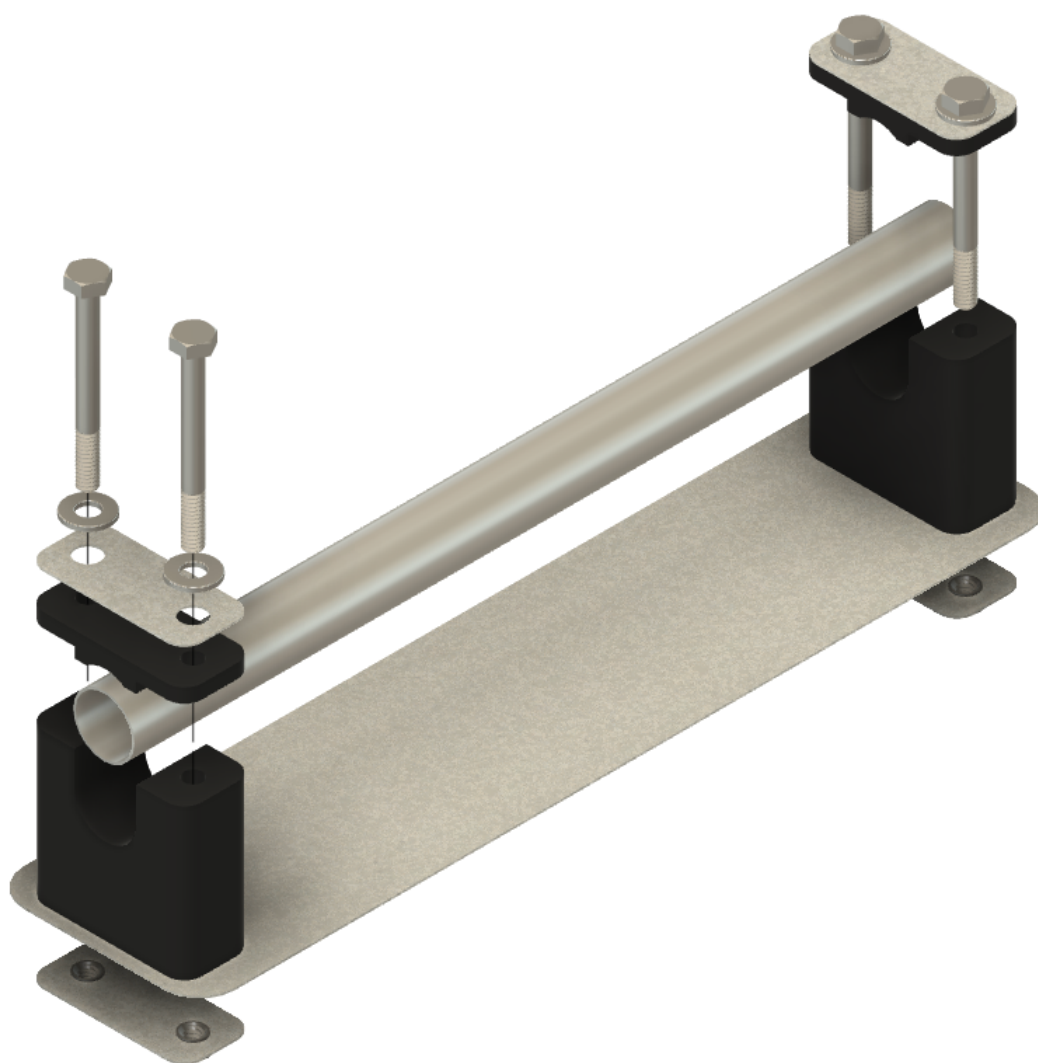


## Section 2: Handle Setup

### ***Step 1: Gather Your Components***

Before beginning, ensure you have the following parts for the handle setup:

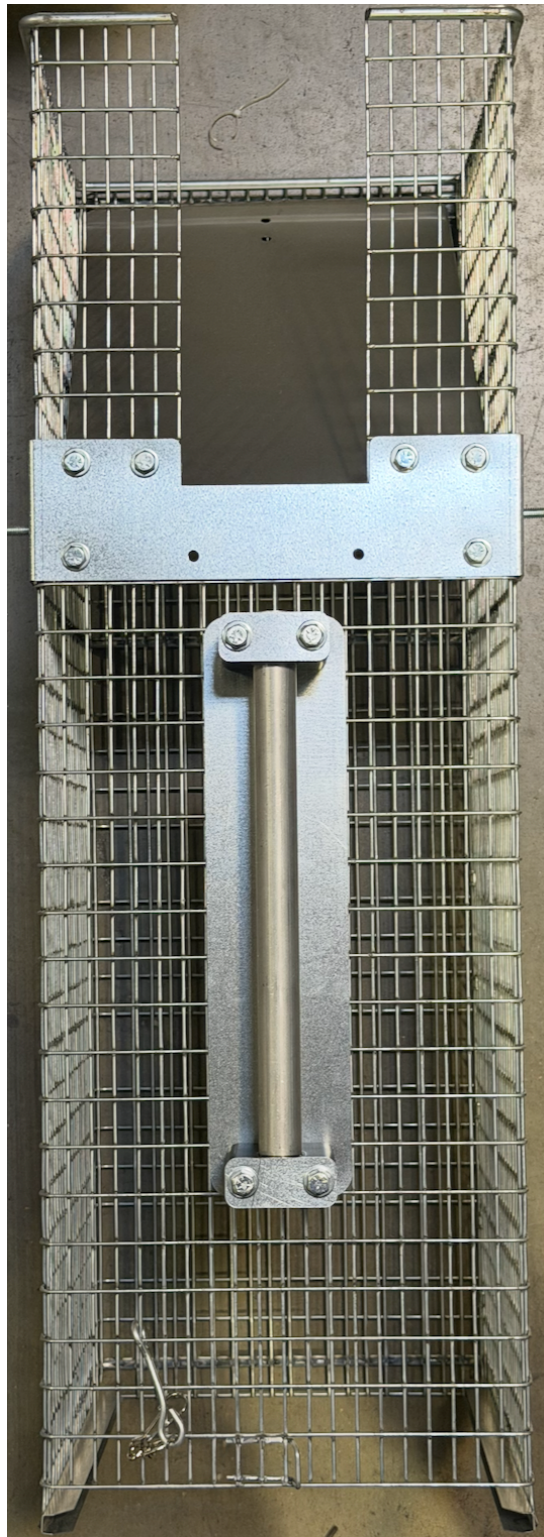
| Quantity | Part                                     |
|----------|------------------------------------------|
| 4        | 3" bolt                                  |
| 1        | Handle Guard                             |
| 1        | Tube                                     |
| 4        | Washers                                  |
| 2        | Top Handle Metal Bracket                 |
| 2        | Upper tube plastic spacer                |
| 2        | Lower tube Plastic Handle Spacer         |
| 2        | Internal Trap Bracket with Press-Fit Nut |
| 1        | Nut Driver                               |



**Step 2: Screw in handle**

1. Center the handle guard on the trap.
2. Insert the two top handle pieces into the holes in the handle guard.
3. Slide the tube between the two handle guards and ensure it is centered.
4. Place the bracket (with the press-fit nut) inside the trap and hand-tighten the bolts.

Do not fully tighten the bolts until the motor connector is fully through the tube



Top View





Inside trap view



### Section 3: POD



#### Step 2: Connect the Wire to the POD

1. Slide the POD down the rails on the trap until it is in place.
2. Attach the clip to the top keep the POD in place



1. Attach the motor's wire to the POD using the locking screw connector, ensuring a secure connection.
2. Use the top latch to secure the pod in place.

Fully tighten all the bolts. At this point, your hardware setup is complete.

If you have issues, check out [Broken link](#)



The OcuTrap hardware now set up!

## Video Assembly

### Unpack the Components

Welcome to the OcuTrap assembly guide! In this guide, we'll walk you through each step of building your OcuTrap to ensure a smooth and successful setup.

Before we begin, make sure you have all the necessary components:

- The OcuTrap Cage
- The POD
- A fully charged battery
- The Motor

*Take a moment to confirm you have all these parts.*

### ***Prepare the Battery***

Ensure the battery is fully charged. Charging typically takes 1-3 hours, so it's best to charge it ahead of time if you haven't done so already.

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

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## Section 1: Door Assembly

### Step 1:

For the motor assembly, gather these components:

- The bracket (top locking mechanism)
- White washers
- Springs
- Nuts
- Assembly tool
- The metal door

### Instructions:

1. Thread the rod through the metal door.
2. On each end of the rod, add a white washer, followed by a spring, and secure it with a nut.

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

### Step 2: Attach the Motor to the Door

To attach the motor to the door, you'll need:

- Two pins
- Two clevises

### Instructions:

1. Align the motor with the metal door attachments.
2. Secure it by inserting a pin at both the top and bottom.
3. Feed the cable through the metal handle.

4. Double-check that all components are securely fastened and that the door moves smoothly.

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

---

## Section 2 : POD Assembly

### Step 1: Connect Battery

1. Connect the yellow connectors together.

### Step 2: Connect the Wire to the POD

1. Connect the wire from the motor to the POD.
2. Make sure the connection is secure.
3. Then mount the pod to the trap

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

Connect the wire to the POD, ensuring the connection is secure.

**With this step, your hardware setup is complete!**

---

## Section 3: Software Setup

1. Create an account on [base.ocutrap.com](https://base.ocutrap.com).
2. Once your account is created, add your trap to it.

**Instructions:**

- Locate the POD for your OcuTrap and open it to find the serial number at the top of the device.
- Log in to your account at [OcuTrap.com](https://OcuTrap.com).
- Navigate to the 'Account' section in the top-right corner of the dashboard, and click **Add Trap** at the bottom of the page.
- Enter the serial number.
- Follow the in-app prompts to enable your subscription.

Once complete, your new trap will appear in your dashboard.

---

**Conclusion**

Congratulations! You've successfully assembled and set up your OcuTrap.

If you have any questions or need assistance, [contact our team](#).

Thank you for choosing OcuTrap, and happy trapping!

**Hardware Features**

Overview of the OcuTrap R1 hardware components and capabilities.

---

***Connectivity***

The OcuTrap connects via **4G LTE cellular** networks, providing nationwide coverage without the need for Wi-Fi or local network setup. Simply power on and the trap connects automatically.

---

***Camera***

The integrated camera provides a clear view inside the trap with:

- **Automatic night vision** — IR LEDs activate in low-light conditions
- **Adjustable image quality** — Choose from 6 resolution sizes
- **Image rotation** — 0°, 90°, 180°, or 270° to match your trap orientation



- **Timelapse photos** — Periodic images while armed (configurable interval)
- 

## ***Door***

The door is powered by a **linear motor** for fast, reliable operation:

- **Close speed**: Less than 0.5 seconds
  - **Open speed**: Less than 1 second
  - **Remote control** — Open, close, and lock via the app
  - **Manual control** — Button sequence on the device
  - **Enhanced closing** — Optional double-close sequence for secure locking
- 

## ***Location***

Track your trap location using the integrated **GPS module**:

- **Satellite positioning** — Displays number of satellites connected
  - **Battery-optimized** — Updates every 8 hours by default
  - **Map view** — See trap location on Map or Satellite view in the app
  - **Automatic updates** — GPS triggered on capture events
- 

## ***Sensors***

OcuTrap includes multiple sensors for monitoring and capture detection:

| Sensor                            | Function                                                          |
|-----------------------------------|-------------------------------------------------------------------|
| <b>Time-of-Flight (ToF)</b>       | Detects animals entering the trap with millimeter precision       |
| <b>Temperature &amp; Humidity</b> | Monitors internal conditions, sends alerts if thresholds exceeded |
| <b>Ambient Light</b>              | Detects darkness to activate night vision automatically           |
| <b>Accelerometer</b>              | Detects tilt and movement of the trap                             |

---

## ***Audio & Haptic Feedback***

- **Buzzer** — Audible beeps for button feedback and status indication
  - **Vibration motor** — Haptic feedback for certain operations
-

## ***Battery***

Rechargeable **12V lithium-ion battery** for extended field deployment:

- **10,000 mAh capacity** — 3+ weeks runtime per charge
  - **5,000 mAh variant** — Available for Canadian shipments (~21 days runtime)
  - **Low battery alerts** — Notifications at 20% and 10% levels
  - **Auto-hibernation** — Protects battery when voltage drops too low
- 

## ***Accessory Port***

**12V accessory port** for powering external devices:

- Connect dispensers, pumps, or other add-on hardware
  - Configurable activation duration (0–30 seconds)
  - Enable/disable via app settings
- 

## ***Smart Detection***

Advanced capture detection with false-trigger prevention:

- **Dual-zone verification** — Objects must pass through detection zone before triggering
  - **Rain/debris filtering** — Rejects oscillating readings from environmental interference
  - **Consecutive reading requirement** — Prevents single-reading false triggers
  - **Pre-capture alerts** — Optional notification when animals approach
- 

## ***Construction***

- **Weather-resistant enclosure** — Built for outdoor field deployment
  - **Dimensions:** 10"W × 12"H × 32"L
  - **Weight:** 24 lbs (10.9 kg)
  - **Target animals:** 5–25 lbs (cats, raccoons, opossums)
- 

For detailed specifications, see [Technical Specifications](#).



## Technical Specifications

This page provides detailed technical specifications for the OcuTrap R1 smart wildlife trap.

---

### Physical Specifications

| Specification         | Value                                                             |
|-----------------------|-------------------------------------------------------------------|
| <b>Dimensions</b>     | 10"W × 12"H × 32"L (25.4cm × 30.5cm × 81.3cm)                     |
| <b>Weight</b>         | 24 lbs (10.9 kg)                                                  |
| <b>Construction</b>   | Weather-resistant enclosure, compatible with Tomahawk trap frames |
| <b>Target Animals</b> | 5–25 lbs (cats, raccoons, opossums, similar wildlife)             |

---

### Door Mechanism

| Specification          | Value                                                             |
|------------------------|-------------------------------------------------------------------|
| <b>Actuator Type</b>   | Linear motor with DRV8873 motor controller                        |
| <b>Close Speed</b>     | < 0.5 seconds                                                     |
| <b>Open Speed</b>      | < 1 second                                                        |
| <b>Door States</b>     | Opening, Closing, Fully Open, Fully Closed, Error, Forced Stopped |
| <b>Control Methods</b> | App remote control, physical button sequence                      |

---

## Battery & Power

| Specification                         | Value                                   |
|---------------------------------------|-----------------------------------------|
| <b>Battery Type</b>                   | Rechargeable KBT 12V Lithium-ion        |
| <b>Standard Capacity</b>              | 10,000 mAh (10Ah) — US shipments        |
| <b>Canadian Variant</b>               | 5,000 mAh (5Ah) — Canadian shipments    |
| <b>Operating Voltage Range</b>        | 7.0V – 15.0V                            |
| <b>Low Battery Warning (20%)</b>      | 10.4V (default, configurable)           |
| <b>Critical Battery Warning (10%)</b> | ~9.5V                                   |
| <b>Auto Power-Off Threshold</b>       | 9.6V (default, configurable 7.0V–12.0V) |
| <b>Runtime (10Ah)</b>                 | 4+ weeks per charge (typical usage)     |
| <b>Runtime (5Ah)</b>                  | ~21 days per charge (typical usage)     |
| <b>Charger (10Ah)</b>                 | 2A @ 12V, ~5–6 hours full charge        |
| <b>Charger (5Ah)</b>                  | 1A @ 12V, ~5–6 hours full charge        |

---

## Connectivity

| Specification              | Value                                      |
|----------------------------|--------------------------------------------|
| <b>Connection</b>          | 4G LTE Cellular                            |
| <b>Coverage</b>            | Multi-network cellular coverage            |
| <b>GPS Update Interval</b> | Every 8 hours (default, battery-optimized) |
| <b>GPS Accuracy</b>        | Minimum 5 satellites, 3D fix required      |

## Sensors

### Time-of-Flight (ToF) Distance Sensor

| Specification           | Value                                      |
|-------------------------|--------------------------------------------|
| <b>Model</b>            | VL53L1X                                    |
| <b>Maximum Range</b>    | 0–4 meters                                 |
| <b>Capture Distance</b> | Configurable 125mm–1000mm (default: 250mm) |
| <b>Detection Zone</b>   | 300–450mm from sensor                      |
| <b>Capture Zone</b>     | 0–250mm from sensor                        |

## Environmental Sensors

| Sensor                 | Function                                 |
|------------------------|------------------------------------------|
| Temperature & Humidity | Environmental monitoring, alerts         |
| Ambient Light          | Automatic day/night detection for camera |
| Accelerometer          | Tilt detection, movement alerts          |

## Temperature Thresholds

| Alert                  | Default Value                           |
|------------------------|-----------------------------------------|
| High Temperature Alert | 45°C (113°F)                            |
| Low Temperature Alert  | -10°C (14°F)                            |
| Alert Interval         | Every 8 hours (configurable 0–48 hours) |

---

## Camera System

| Specification            | Value                                         |
|--------------------------|-----------------------------------------------|
| Night Vision             | Integrated IR LED (automatic activation)      |
| IR Brightness            | 0–100% (configurable min/max)                 |
| Image Sizes              | QVGA to UXGA (6 selectable sizes)             |
| Rotation Options         | 0°, 90°, 180°, 270°                           |
| Color Modes              | Grayscale (dark conditions), Color (daylight) |
| Dark Detection Threshold | 25 lux (default, configurable 1–100 lux)      |
| Timelapse Interval       | 0–24 hours (default: 6 hours)                 |

---

## Capture Detection System

OcuTrap uses a sophisticated dual-zone verification system to prevent false triggers from rain, debris, or non-target movement.

### Detection Process

1. **Object enters Detection Zone** (300–450mm) → 3+ consecutive valid readings required
2. **Object verified** → Pre-capture alert sent (if enabled)
3. **Object enters Capture Zone** (0–250mm) → 3+ consecutive readings trigger capture

4. **Door closes** → Capture photo taken and transmitted

### False Trigger Prevention

- **Signal quality filtering** — Validates signal-to-ambient ratio (optimized for outdoor sunlight)
  - **Distance consistency checks** —  $\pm 20\text{mm}$  tolerance rejects oscillating readings
  - **Rain detection** — Oscillation patterns identified and filtered
  - **Status validation** — Only Status 0 (valid measurement) readings accepted
- 

### What's in the Box

- OcuTrap R1 Smart Cage Unit
  - 12V Lithium-ion Battery (10Ah US / 5Ah Canada)
  - Battery Charger (2A or 1A depending on battery)
  - Quick-Start Guide
  - Assembly hardware
- 

### Environmental Ratings

| Specification         | Value                                                      |
|-----------------------|------------------------------------------------------------|
| Operating Temperature | -10°C to 45°C (14°F to 113°F)                              |
| Weather Resistance    | Designed for outdoor field deployment                      |
| Recommended Placement | Areas with strong cellular signal for optimal battery life |

---

### Firmware & Software

| Specification | Value              |
|---------------|--------------------|
| Updates       | Over-the-air (OTA) |
| Mobile App    | iOS and Android    |

---

## Regulatory Information

For warranty, safety, and compliance information, see:

- [Safety Information](#)
- [Warranty Information](#)
- [Legal Disclaimers](#)

## App

### Creating an account

<https://base.ocutrap.com/signuplogin>

### Mobile app

#### Apple Devices



## **Android Devices**



### **Adding a trap to your account**

You need to be logged into OcuTrap.com and have the POD with you in order to register.

#### **First-time registration**

You need to be logged in to OcuTrap.com and have the POD with you in order to register.

1. Locate the POD for the OcuTrap you wish to register.
2. Open the POD and locate the serial number on the top of the device.
3. On your device, account -> add trap at the bottom of the page.
4. You will be prompted to enter the serial number.
5. Once you complete the process, the new trap will appear in your dashboard



**Graham Patterson**

Manage your OcuTrap account information such as name, email, phone number, password and your payment method.

**Personal**

**Name**  
Graham Patterson

[Update](#)

**Email**  
graham@ocutrap.com

[Update](#)

**Phone Number**  
(512) 968-1348

[Update](#)

**Security**

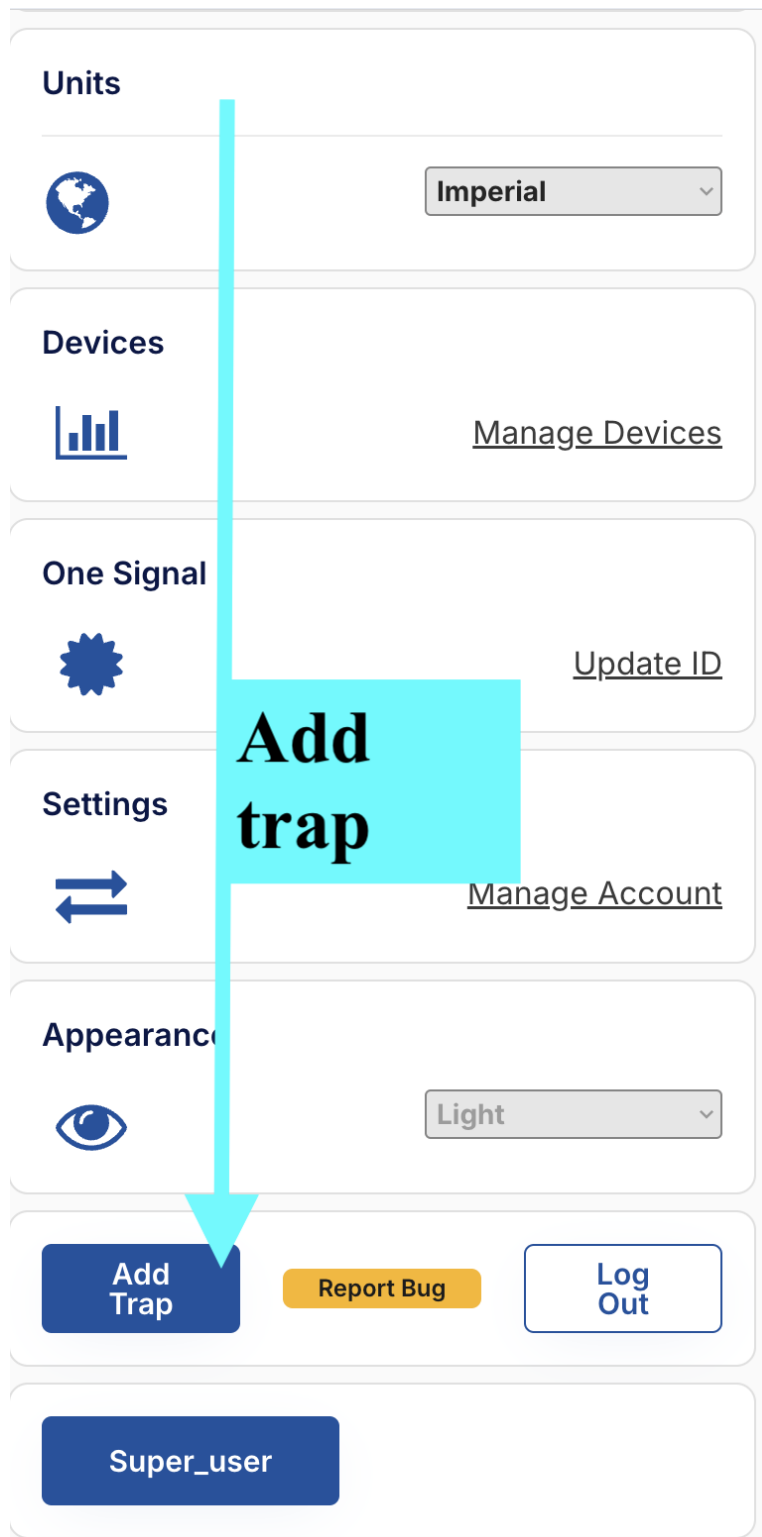
**Password**

[Update](#)

**Billing**

[Billing Page](#)

**Scroll  
Down**



## Billing

[billing](#)



## Open/ Closed Button

*Make sure you are logged in to your account and on the traps page.*

### ***Open/ Closed Button***

- **Functionality:** Allows controlling the trap's door.
- **Open:** Lifts the door for resetting or letting animals exit.
- **Close:** Shuts and locks the door preventing escape or for safe transportation.
- **Subtext:** Shows the latest reported door state.

## Arm/ Un-arm Button

*Make sure you are logged in to your account and on the traps page.*

### ***Armed State***

- **Purpose:** The trap is set to capture an animal.
- **Safety Requirement:** The user must manually open the door to activate this state. This ensures safety during the arming process.
- **Behavior:** Once armed and the door is confirmed open, the trap enters a low-power mode, conserving energy while remaining active for an animal to enter.
- **Notifications:** The trap maintains a connection by sending periodic updates. If disconnected for more than an hour, a notification is sent to the user.

### ***Scouting State***

- **Purpose:** The trap is set for observing activity without closing the door.
- **Safety Requirement:** The door must already be fully open before Scouting Mode can be enabled.
- **Behavior:** Scouting uses the same animal-detection logic as armed mode, including the outer pre-capture zone and the final trigger-distance zone, but it never closes the door.
- **Notifications:** The user can receive an alert when an animal first approaches and another alert if it reaches the trigger distance. After the

animal leaves, scouting resets and waits about 5 minutes before sending a new set of scouting alerts.

For full details, see the [Scouting Mode](#) page.

### **Unarmed State**

- **Purpose:** The trap is not set to capture and is in a standby mode.
- **Behavior:** In this state, the trap listens for commands and does not initiate the door closure if motion is detected.
- **Transition:** After capturing an animal, the trap automatically shifts to this state, signaling the door is securely closed.

### **Scouting Mode**

Scouting Mode lets you observe trap activity without ever closing the door — useful for confirming the right animals are visiting before you commit to a real capture.

#### **How it differs from Armed mode**

|                                 | <b>Armed</b>                       | <b>Scouting</b>                                                          |
|---------------------------------|------------------------------------|--------------------------------------------------------------------------|
| <b>Detects animals</b>          | Yes                                | Yes (same logic as Armed)                                                |
| <b>Closes the door</b>          | Yes — on first verified detection  | <b>Never</b> — observation only                                          |
| <b>Sends pre-capture alerts</b> | Yes (if enabled)                   | Yes                                                                      |
| <b>Captures images</b>          | Yes — at zone entry and at trigger | Yes — on detection, then about every 60 s while in zone (15 s – 5 min)   |
| <b>Reset after activity</b>     | N/A (door closes; trap captured)   | Auto-reset ~5 min after departure is confirmed (~30 s of clear readings) |

#### **Activating Scouting Mode (walkthrough)**

**Open / Close / Arm** are on the trap card. **Scout** lives one tap away inside that card's **Controls** popup.

1. Sign in to the OcuTrap app at [app.ocutrap.com](http://app.ocutrap.com) (or the mobile app). You'll land on your trap list.
2. Find the trap you want to scout. Each card shows its status, battery %, and a green dot if online.

3. Confirm the status reads **Open / Unarmed**. If the door is closed, tap **Open** on the trap card first and wait for the status to update.
4. Tap **Controls** on the trap card. A popup opens with a top row of icons: **GPS, Data, Buzzer, Reboot, Hibernation, Scout**.
5. Tap **Scout** (rightmost, eye icon). The trap runs an obstruction check — if anything is in the capture zone, scouting is refused so you can clear it and retry.
6. Once the check passes, the card status flips to **Scouting** and the popup icon flips to **Stop Scout** (eye with slash).

**Tip:** If **Scout** seems unresponsive, the door usually isn't fully open, the trap is offline, or **Last heard** at the bottom of the popup is stale. Wait for the next check-in or re-open the door.

### What you'll see while scouting

- **Pre-capture alerts** when an animal enters the outer detection zone (if enabled).
- **Trigger alerts** when an animal reaches the trigger distance. The trap takes a photo but the door **does not close**.
- **Periodic photos** about every 60 seconds (configurable 15 s – 5 min) while the animal stays in zone. The trap waits for each photo to finish before starting the next, and saves the closest approach frame if the camera is busy. This draws more power than idle scouting; if battery matters, ask your fleet admin to lengthen the interval.
- **Cooldown after departure** — the trap waits ~30 seconds of clear readings to confirm the animal has actually left, then another ~5 minutes before re-arming the alerts. Expect about 5.5 minutes of quiet after the last in-zone reading.

### Exiting Scouting Mode

1. **Stop Scout** in the Controls popup — returns the trap to Unarmed, door still open.
2. **Close** on the trap card — closes the door and returns to Unarmed in one step.

### When Scouting Mode isn't what you want

- **Capturing a confirmed animal** — tap **Arm** on the trap card.

- **A one-off photo** — open Controls and tap **Request Image** (doesn't change trap state).
- **Testing the door mechanism** — use **Open** / **Close** on the trap card.

## Troubleshooting

- **"Refused with an obstruction error"** — clear the capture zone, then retry.
- **"Refused with a door-not-open error"** — tap **Open** on the trap card, wait for the status to read **Open**, then retry **Scout**.
- **Repeated alerts from the same animal** — expected if the animal is cycling in and out of detection zones faster than the ~5.5 minute cooldown can reset.

## Trap Settings

Manage: 6255671

[Change Trap Name](#)

Type trap notes here.

---

**Notifications**

|         |                   |
|---------|-------------------|
| Error   | Push-notification |
| Alert   | Push-notification |
| Capture | Push-notification |

---

**Trigger Distance** 9 in

*Set trigger distance*

**Image Size** Medium

*Change image size*

[Device info](#) [More Settings](#)

The trap settings are used to set notifications, trigger distance in armed mode set other sizes. For more options click more settings.

## Settings Reference

This page provides a complete reference of all configurable settings on your OcuTrap, including their ranges, defaults, and descriptions.

**Note:** After changing settings, a reboot of the trap is recommended to ensure all changes are applied correctly.

---

### *Capture & Detection Settings*

| Setting                   | Range                   | Default          | Description                                                                                                            |
|---------------------------|-------------------------|------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Capture Distance</b>   | 125–1000mm<br>(5–39 in) | 250mm<br>(10 in) | Distance from sensor that triggers a capture. Objects crossing this threshold will close the door.                     |
| <b>Pre-Capture Alerts</b> | On/Off                  | On               | Sends an alert when an animal enters the detection zone (before capture). Includes a 2-minute cooldown between alerts. |

### *Detection Zones Explained*

OcuTrap uses two detection zones to verify captures and reduce false triggers:

- **Detection Zone:** 300–450mm from sensor — Object must show 3+ consecutive valid readings here first
- **Capture Zone:** 0–250mm from sensor — Final trigger point that closes the door

This dual-zone system prevents false triggers from rain, debris, or quick movements.

---

## Camera Settings

| Setting                      | Range               | Default | Description                                                                                           |
|------------------------------|---------------------|---------|-------------------------------------------------------------------------------------------------------|
| <b>Camera Time Lapse</b>     | 0–24 hours          | 6 hours | How often the camera takes periodic photos while armed. Set to 0 to disable timelapse.                |
| <b>Camera Quality</b>        | 1–6                 | 2       | Image resolution size (1=QVGA smallest, 6=UXGA largest). Higher = better quality but more data/power. |
| <b>Rotate Image</b>          | 0°, 90°, 180°, 270° | 0°      | Rotates captured images. Useful if trap is mounted in non-standard orientation.                       |
| <b>Dark Lux Threshold</b>    | 1–100 lux           | 25 lux  | Light level below which the environment is considered "dark" and IR lighting activates.               |
| <b>Minimum IR Brightness</b> | 0–100%              | 10%     | Minimum infrared LED brightness in dark conditions.                                                   |
| <b>Maximum IR Brightness</b> | 0–100%              | 100%    | Maximum infrared LED brightness. Lower values reduce glare and save power.                            |

## Camera Quality Levels

| Level | Resolution       | Best For                                    |
|-------|------------------|---------------------------------------------|
| 1     | QVGA (320×240)   | Fastest transfer, lowest data usage         |
| 2     | VGA (640×480)    | Good balance of quality and speed (default) |
| 3     | SVGA (800×600)   | Better detail                               |
| 4     | XGA (1024×768)   | High detail                                 |
| 5     | SXGA (1280×1024) | Very high detail                            |
| 6     | UXGA (1600×1200) | Maximum detail, highest data usage          |

## Battery & Power Settings

| Setting                  | Range      | Default          | Description                                                      |
|--------------------------|------------|------------------|------------------------------------------------------------------|
| <b>Battery Type</b>      | 5Ah / 10Ah | Varies by region | Must match installed battery for accurate level estimation.      |
| <b>Battery Alerts</b>    | On/Off     | On               | Enables low battery notifications at 20% and 10% levels.         |
| <b>Power-Off Voltage</b> | 7.0–12.0V  | 9.6V             | Voltage threshold below which the trap automatically hibernates. |

## Battery Alert Thresholds

| Alert Level     | Default Voltage | Description                                              |
|-----------------|-----------------|----------------------------------------------------------|
| 20% Warning     | 10.4V           | Low battery warning sent                                 |
| 10% Critical    | ~9.5V           | Critical battery warning sent                            |
| Auto Power-Off  | 9.6V            | Trap enters hibernation to protect battery               |
| Reset Threshold | 11.0V           | Battery must reach this level to clear low-battery flags |

## Temperature Alert Settings

| Setting                    | Range        | Default      | Description                                                                 |
|----------------------------|--------------|--------------|-----------------------------------------------------------------------------|
| Temperature Alerts         | On/Off       | On           | Enables alerts when temperature exceeds thresholds.                         |
| High Temperature Limit     | Configurable | 45°C (113°F) | Alert sent if internal temperature rises above this.                        |
| Low Temperature Limit      | Configurable | -10°C (14°F) | Alert sent if internal temperature falls below this.                        |
| Temperature Alert Interval | 0–48 hours   | 8 hours      | Minimum time between temperature alerts. Set to 0 to disable repeat alerts. |

## GPS & Location Settings

| Setting        | Range        | Default | Description                                                                 |
|----------------|--------------|---------|-----------------------------------------------------------------------------|
| Location (GPS) | On/Off       | On      | Enables periodic GPS location updates.                                      |
| GPS Interval   | Configurable | 8 hours | How often the trap updates its GPS position. Longer intervals save battery. |

## GPS Behavior Details

- **First fix delay:** 5-minute initial delay after boot before first GPS acquisition
- **Acquisition timeout:** 3 minutes for first fix, 2 minutes for subsequent fixes
- **Fix requirements:** Minimum 5 satellites, 3D fix required for valid position
- **Automatic updates:** GPS automatically triggered on capture events

## Accessory Port Settings

| Setting                 | Range      | Default | Description                                                 |
|-------------------------|------------|---------|-------------------------------------------------------------|
| <b>Accessory</b>        | On/Off     | Off     | Enables the 12V accessory port for external devices.        |
| <b>Accessory Timing</b> | 0–30,000ms | —       | Duration the accessory port remains powered when activated. |

The 12V accessory port can power external devices like dispensers, pumps, or other add-on hardware.

---

## Capture Alert Settings

| Setting                        | Range      | Default | Description                                                                                  |
|--------------------------------|------------|---------|----------------------------------------------------------------------------------------------|
| <b>Capture Alerts Interval</b> | 0–48 hours | 8 hours | Minimum time between capture alert notifications. Prevents repeated alerts for same capture. |

---

## Hardware & Feedback Settings

| Setting                      | Options         | Default  | Description                                                          |
|------------------------------|-----------------|----------|----------------------------------------------------------------------|
| <b>User Beeps</b>            | On/Off          | On       | Audible beeps for button presses, state changes, and feedback.       |
| <b>Enhanced Door Closing</b> | On/Off          | Off      | Performs additional open/close cycle to ensure door is fully locked. |
| <b>Units</b>                 | Metric/Imperial | Imperial | Display units for distance and temperature throughout the app.       |

---

## Image Cropping Settings

These settings remove portions of the image before processing. Values are percentages of the image dimension.

| Setting            | Range | Default | Description                     |
|--------------------|-------|---------|---------------------------------|
| <b>Left Crop</b>   | 0–50% | 0%      | Removes left portion of image   |
| <b>Right Crop</b>  | 0–50% | 0%      | Removes right portion of image  |
| <b>Top Crop</b>    | 0–50% | 0%      | Removes top portion of image    |
| <b>Bottom Crop</b> | 0–50% | 0%      | Removes bottom portion of image |

---



## Settings Locations

Settings are accessible in different locations:

| Location                                  | Settings Available                 |
|-------------------------------------------|------------------------------------|
| App → Trap → Settings → More Settings     | Most user settings                 |
| App → Trap → Settings → Advanced Settings | Temperature alerts, image cropping |
| App → Account                             | Notification preferences, units    |

## Tips for Optimal Settings

### *For Maximum Battery Life*

- Set GPS Interval to 8+ hours
- Use Camera Quality level 1–2
- Set Camera Time Lapse to 6+ hours or disable
- Reduce Maximum IR Brightness if images are overexposed

### *For Best Image Quality*

- Use Camera Quality level 4–6
- Adjust Dark Lux Threshold based on your deployment (lower = earlier IR activation)
- Fine-tune IR brightness settings for your environment

### *For Fastest Response*

- Keep GPS enabled for accurate location on captures
- Enable Pre-Capture Alerts to see animals approaching
- Use shorter Capture Alert Intervals if monitoring actively

## More Settings Overview

**Note:** After changing any settings, a reboot of the trap is recommended to ensure all changes are applied correctly.

### *Battery Type*

Selects the battery configuration installed in the trap. This setting is used to improve battery level estimation, charging behavior, and low battery alerts. Always match this setting to the actual battery installed.

---

### *Accessory*

Enables or disables the accessory port on the trap. When enabled, the accessory port can power external devices such as dispensers or add-on hardware.

---

### *Accessory Timing*

Controls how long the accessory port remains powered when activated. Shorter durations reduce power consumption, while longer durations may be required for certain accessories.

---

### *User Beeps*

Enables or disables audible beeps from the trap hardware. Beeps are typically used for user feedback during actions such as button presses, state changes, or troubleshooting.

---

### *Enhanced Door Closing*

Enables an enhanced door close and re-lock sequence. When enabled, the trap will perform an additional open and close cycle to help ensure the door is fully closed and locked.

---

### *Camera Time Lapse*

Sets how often the camera captures periodic photos while the trap is armed. More frequent photos provide better visibility but increase power and data usage.

---

### *Camera Quality*

Controls the image quality of photos taken by the trap camera. Higher quality images improve clarity but increase capture time, data usage, and power consumption.

---

### *Rotate Image*

Adjusts the orientation of captured images. This is useful if the trap is mounted in a non-standard orientation.

---

### *Dark Lux Threshold*

Defines the ambient light level at which the system considers the environment to be dark. This threshold is used to determine when infrared lighting should be activated.

---

### *Minimum IR Brightness*

Sets the minimum brightness level for the infrared LEDs. This ensures a baseline level of illumination in dark environments.

---

### *Maximum IR Brightness*

Sets the maximum brightness level for the infrared LEDs. Limiting the maximum brightness can help reduce glare, reflections, and power usage.

---

### *Battery Alerts*

Enables or disables low battery alerts. When enabled, the system will notify users when the battery voltage drops below a defined threshold.

---

### *Pre-Capture Alerts*

Enables or disables alerts that occur before a capture event. These alerts can provide early notification of activity near or inside the trap.

---

### *GPS Interval*

Sets how often the trap updates its GPS location. Shorter intervals provide more frequent location updates but increase power consumption.

---

### *Location*

Enables or disables periodic location logging. When enabled, the trap logs its location at regular intervals for tracking and history purposes.

---

## **Advanced Settings**

**Note:** After changing any advanced settings, a reboot of the trap is recommended to ensure all changes are applied correctly.

---

## *Temperature Alerts*

Enables or disables alerts when the trap temperature goes outside the configured range. When enabled, the trap will periodically evaluate temperature readings and send alerts if thresholds are exceeded.

The selected interval determines how often temperature alerts can be sent.

---

### *Temperature Alert Thresholds*

Defines the temperature limits that trigger alerts. These values use the selected unit system, shown as Imperial in the interface.

#### **Above Upper Limit**

Sets the maximum allowable temperature. An alert is sent if the internal trap temperature rises above this value.

#### **Below Lower Limit**

Sets the minimum allowable temperature. An alert is sent if the internal trap temperature falls below this value.

---

### *Capture Alerts*

Controls notifications related to capture events.

#### **Capture Alerts Interval**

Defines the minimum amount of time between capture alerts. This helps prevent repeated notifications for the same capture event or ongoing activity.

---

## *Image Cropping*

Adjusts how images are cropped before being processed or uploaded. Cropping can be used to remove unnecessary areas of the image, reduce file size, or focus on a specific region inside the trap.

### **Left Crop**

Removes a portion of the image from the left side.

### **Right Crop**

Removes a portion of the image from the right side.

### **Top Crop**

Removes a portion of the image from the top.

### **Bottom Crop**

Removes a portion of the image from the bottom.

---

## **Notification Settings**

- **Error**  
Triggered when a trap encounters an error or fails to operate as expected.
- **Alert**  
Activated to provide warnings or important updates regarding a trap's performance.
- **Capture**  
Notifies you when a trap successfully captures or detects the intended target.

### ***Notification Options***

When configuring notifications, you can choose from the following delivery options:

- **No Notification**  
No alert will be sent.

- **Email**

An email will be sent to your registered email address.

- **Push Notification**

A push notification will be sent to your mobile app (if installed).

- **Push Notification + Email**

Both a push notification and an email will be sent.

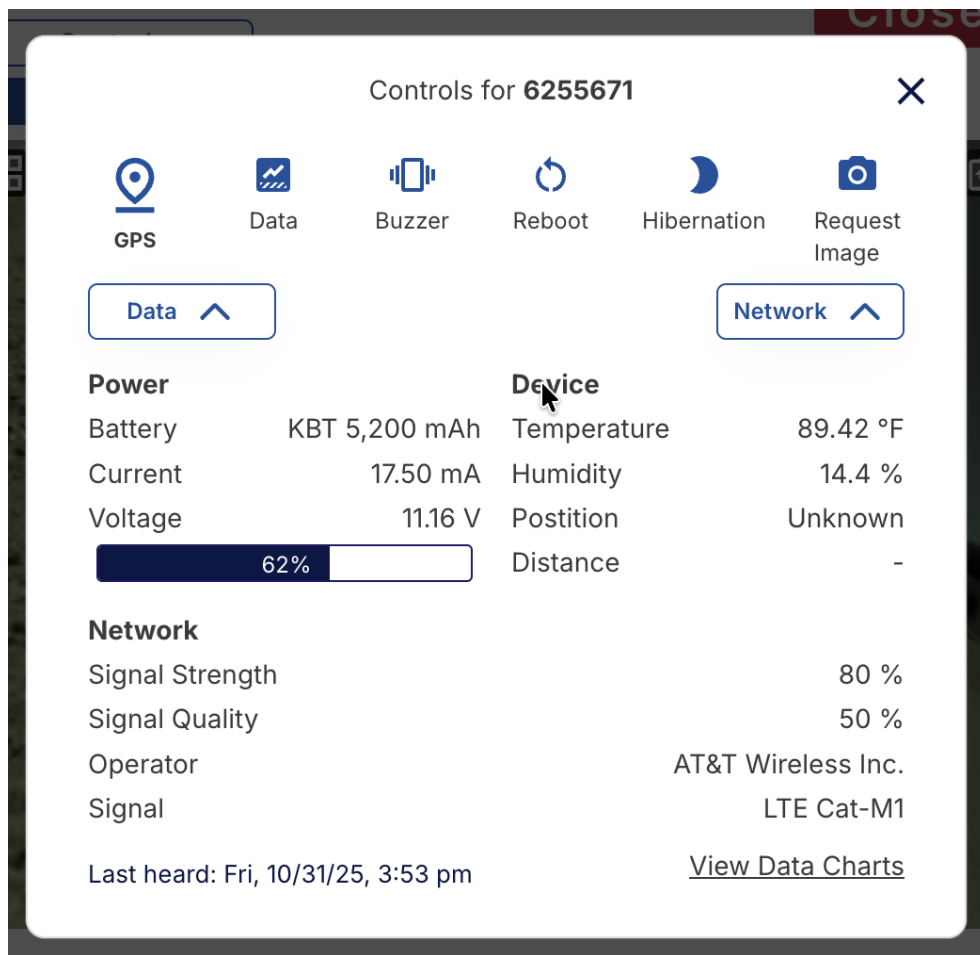
### ***How to Configure Notification Settings***

To configure the notification settings for your traps:

1. Open your OcuTrap dashboard.
2. Locate the trap you wish to modify.
3. Click the **Settings** button for that trap.
4. Select your preferred option for each notification type (Error, Alert, Capture) from the drop-down menu.

Ensure that your mobile app is installed and properly configured if you choose the push notification option.

## Trap Control



### Power

- **Battery Type & Voltage:** Stay informed about the trap's power source, including battery type and voltage, for consistent energy monitoring.

### Device

- **Firmware & Temperature**

Shows the device's firmware version and temperature, aiding in effective management and monitoring.

### Network

- **Signal Quality:** Provides an evaluation of the communication link's integrity.



- **Network Operator:** Identifies the carrier facilitating the trap's connection.
- **Signal Strength:** Measures the power level received by the trap.
- **Last Activity:** Marks the most recent timestamp of the trap's network communication.

## ***Data Charts***

Visual charts provide graphical analysis of the trap's performance over time.

## ***Remote Actions***

Interact directly by:

- Requesting latest data
- Sending a buzz
- Rebooting the device
- Hibernation (go to sleep until battery change or power button press)

## **Logs**

Logs are essential tools for both owners and management users, offering a detailed history of past actions and errors associated with traps. They serve as invaluable resources for troubleshooting, monitoring trap performance, and ensuring accountability.

## **Deleting an Image**

### **1. Open the image gallery**

- From the trap dashboard, scroll to the image gallery below the live image.
- Use the left and right arrows to navigate between images if needed.

### **2. Select the image to delete**

- Click or tap on the image thumbnail you want to remove.
- The image will open in a popup with the image view.

### **3. Click the delete icon**

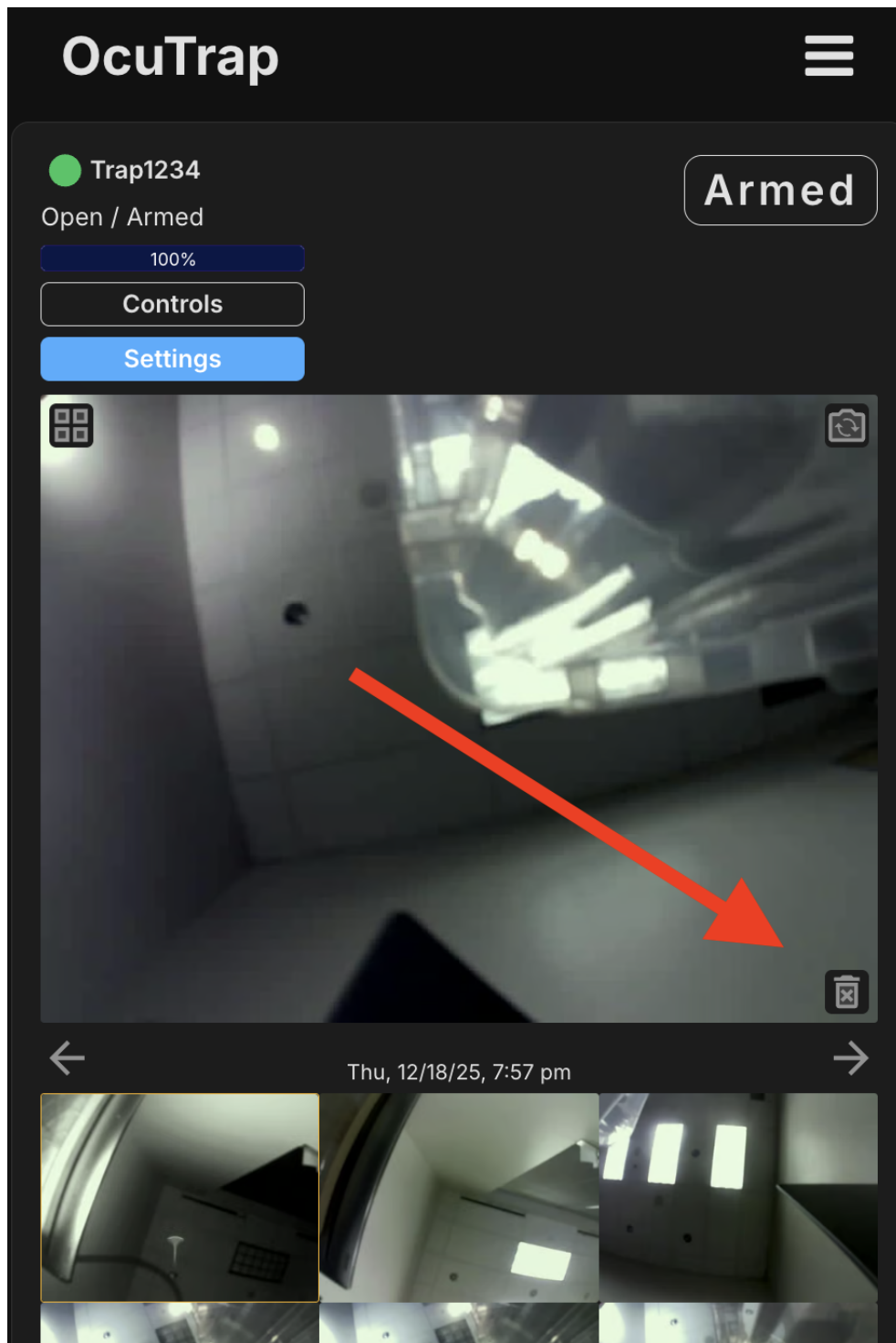
- In the bottom right corner of the image view, click the trash can icon.
- This will open a confirmation dialog.

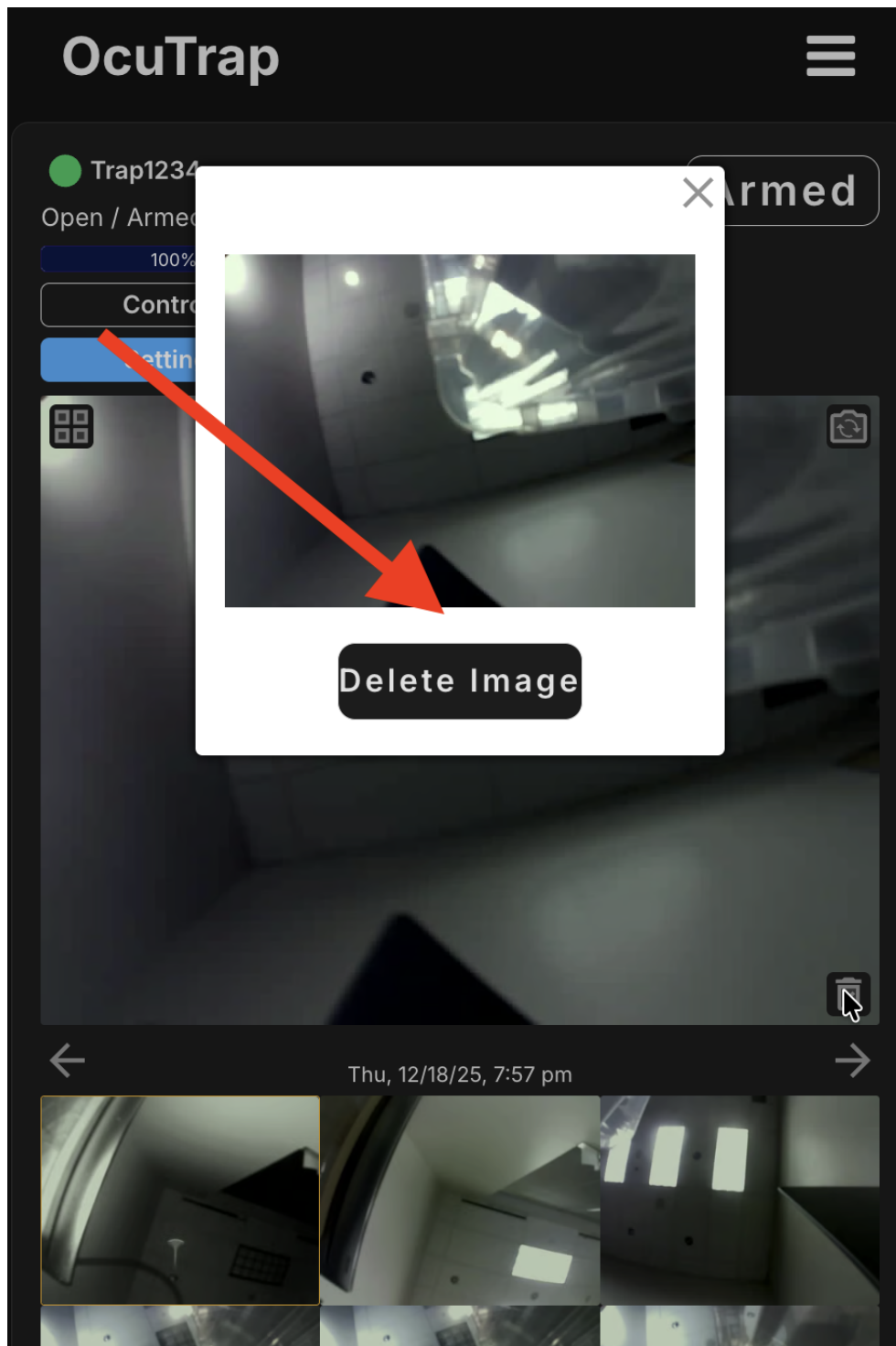
#### 4. Confirm deletion

- In the confirmation window, click **Delete Image**.
- The image will be permanently removed from the trap record.

#### *Important Notes*

- Image deletion is **permanent** and cannot be undone.
- Deleting an image does not affect trap operation, settings, or status.
- If you do not see the delete icon, ensure you have the correct permissions and that the image has fully loaded.





### *Troubleshooting*

- **Delete button does not appear**
  - Refresh the page and reopen the image.
  - Verify your internet connection.

- **Image does not disappear after deletion**

- Refresh the gallery view.
- Allow a few seconds for the change to sync.

For additional help, refer to the general Images and Gallery documentation or contact OcuTrap support.

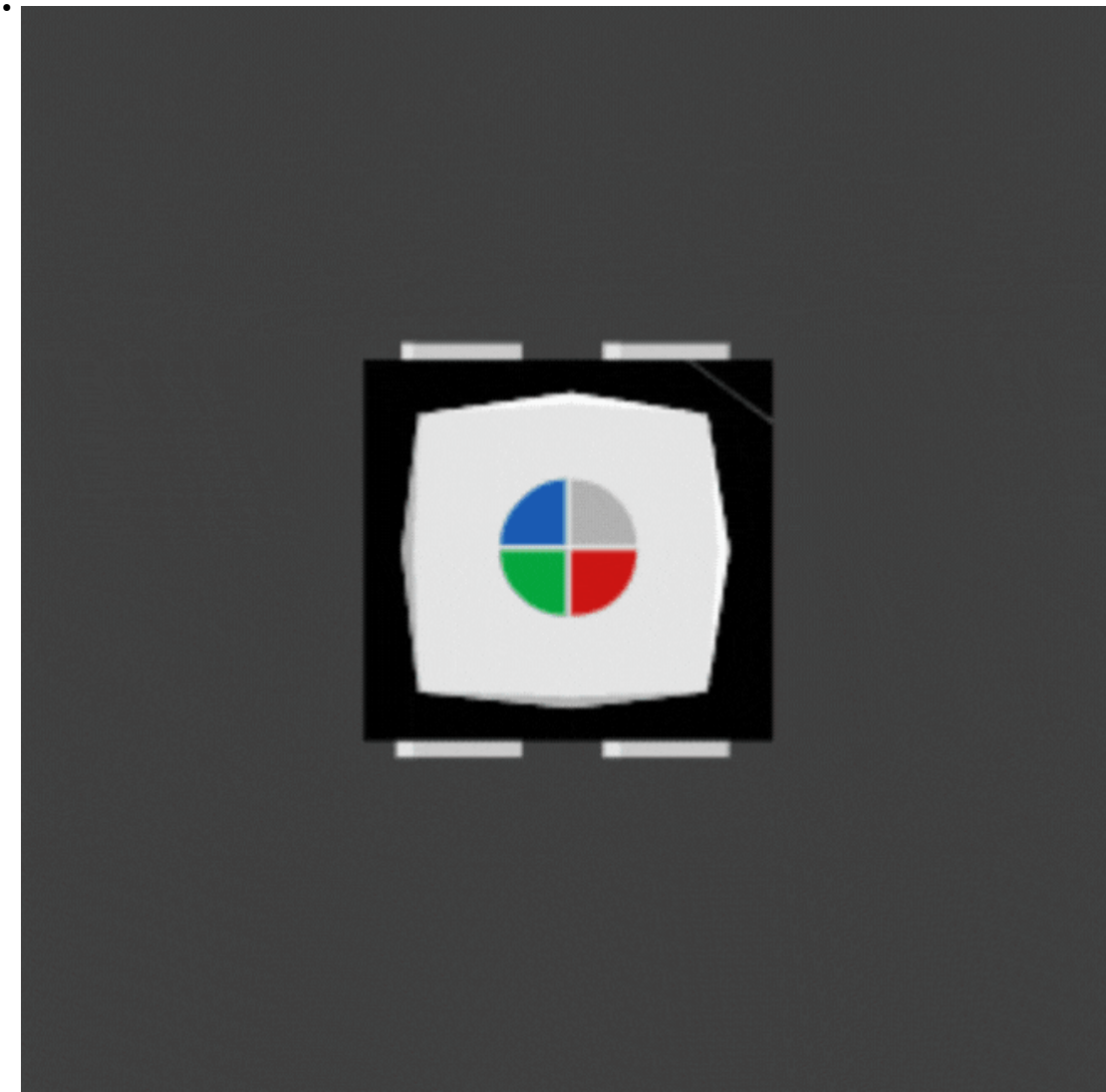
## **Resetting Password**

If you already have an account, go to <https://base.ocutrap.com/> click "forgot password".

## **LED modes**

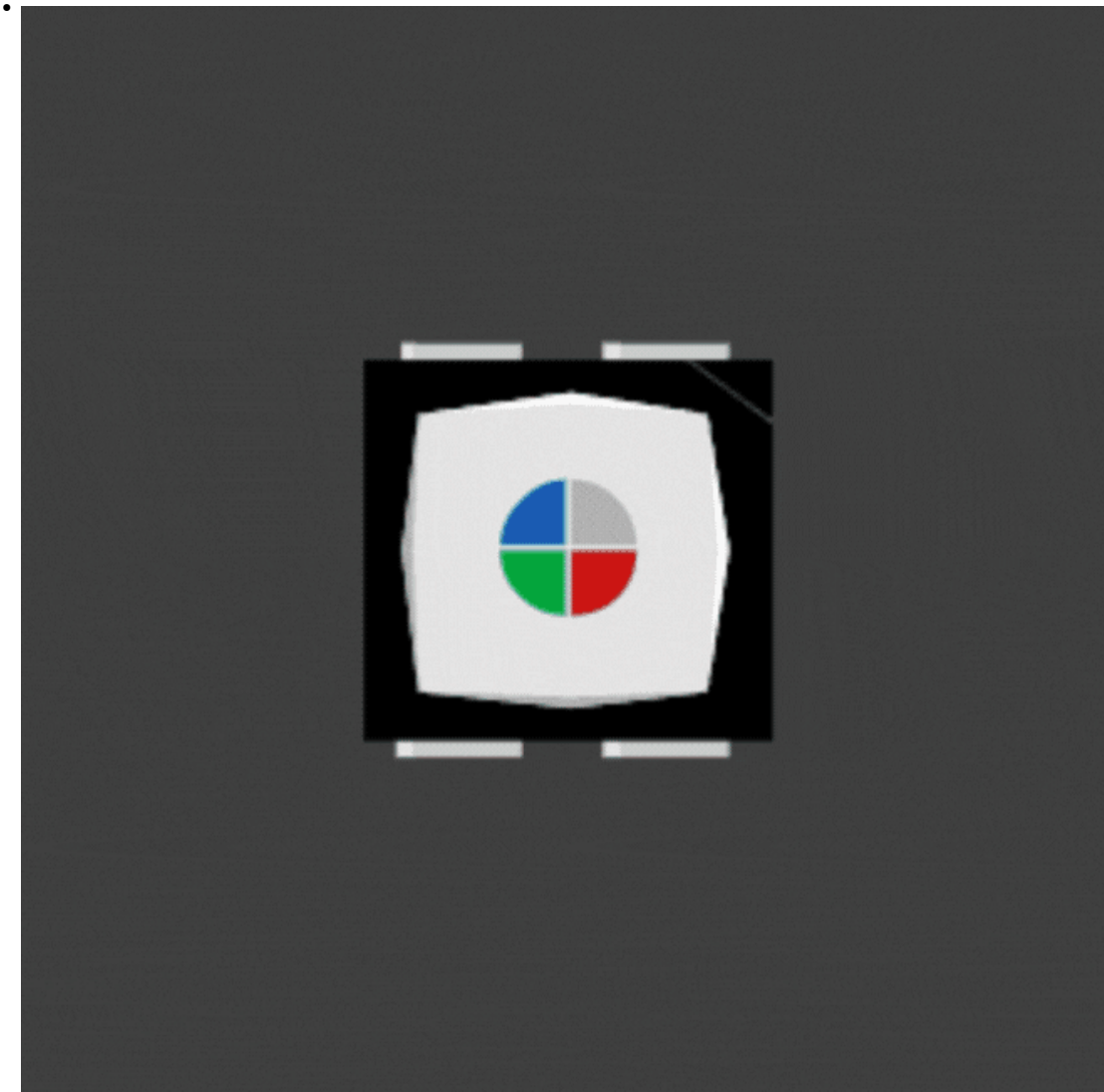
### ***Connected Mode***

- **Breathing Cyan:** The device is successfully connected to the Internet, enabling function calls and code flashing.



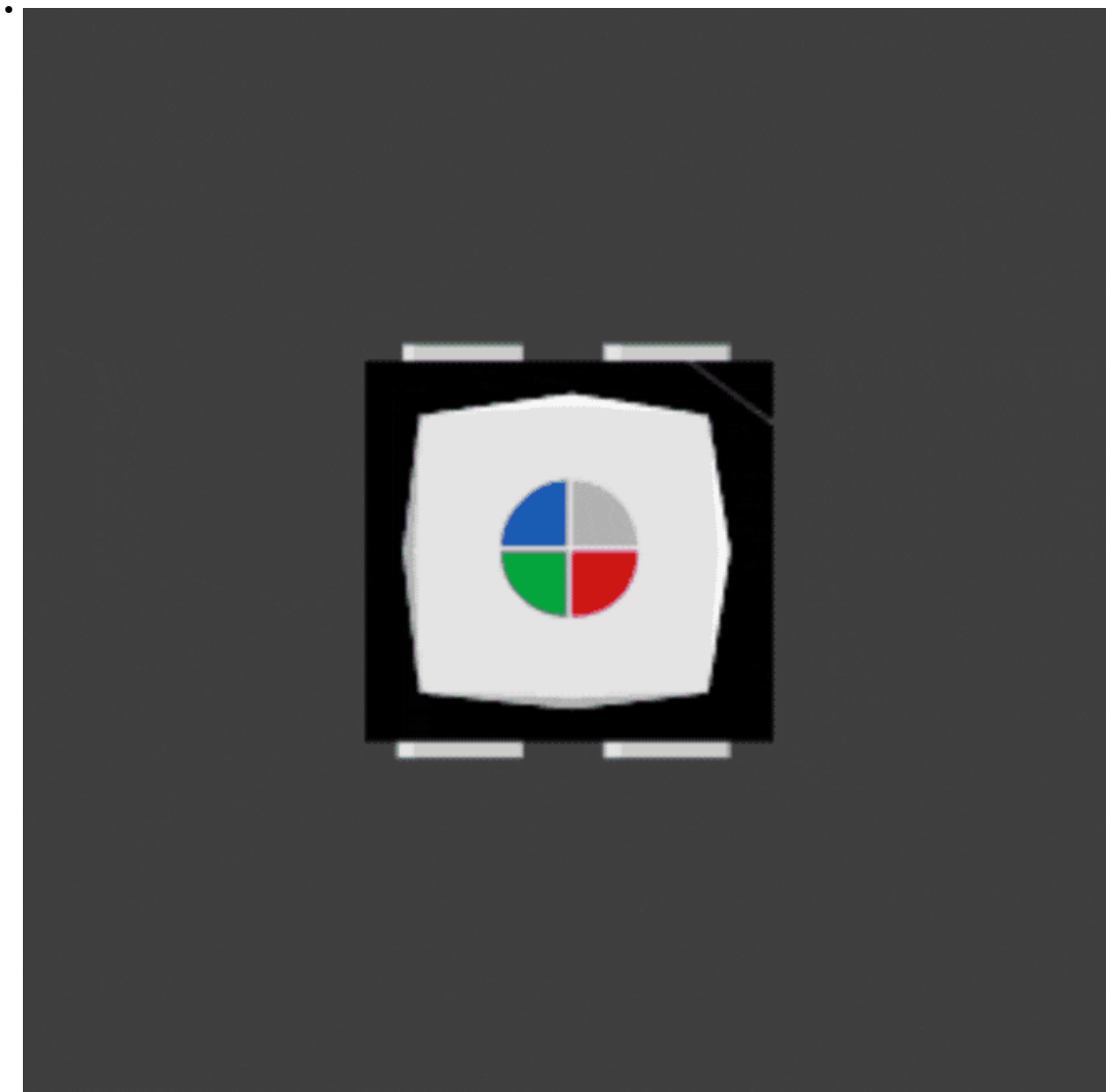
### ***Connecting to the Cloud***

- **Rapidly Blinking Cyan:** Signifies the process of cloud connection, usually following a green blink indicating cellular connection attempts.



### ***OTA Firmware Update***

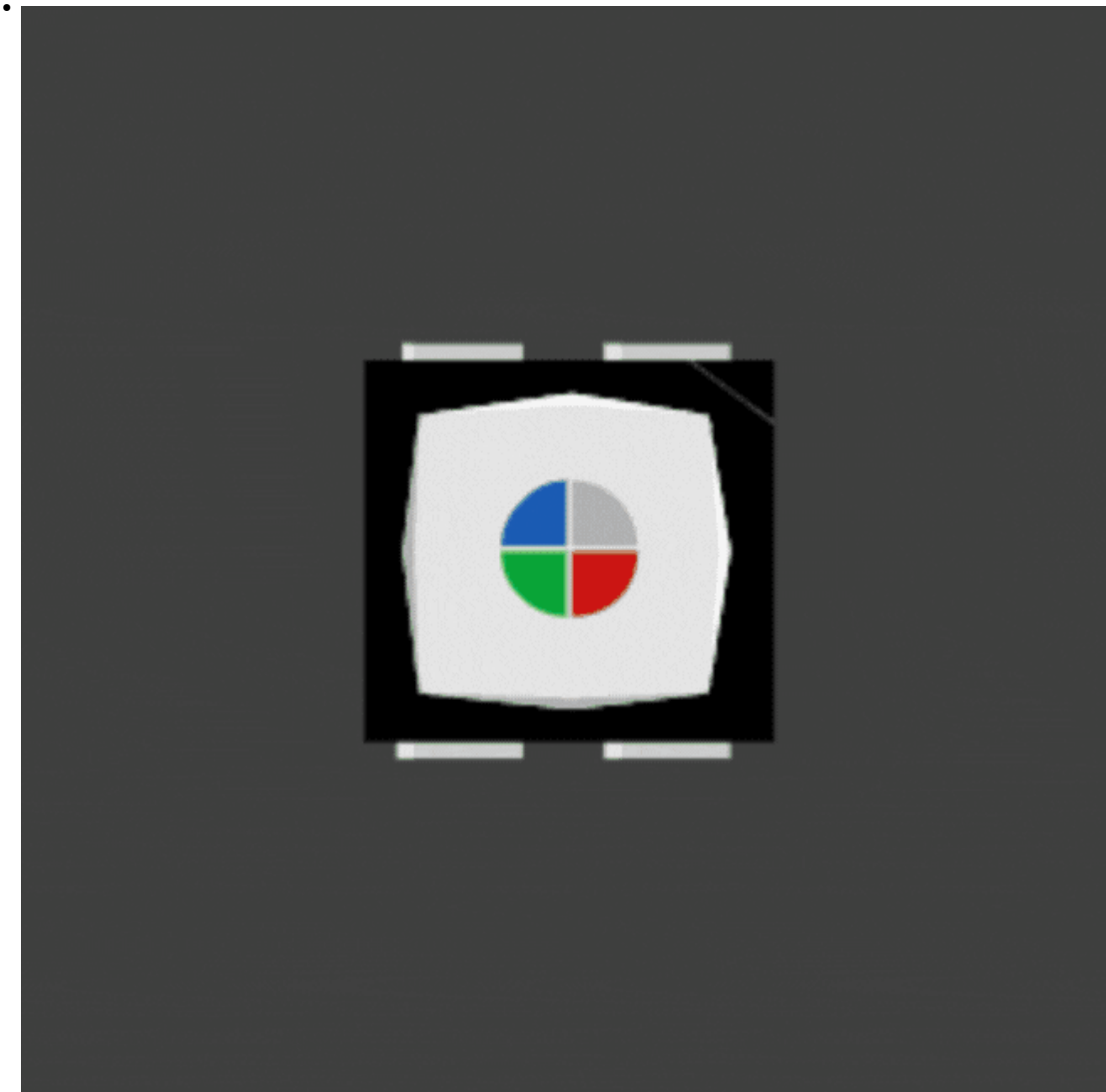
- **Blinking Magenta:** Indicates ongoing firmware updates or code flashing. Entering this mode via MODE button during boot signals an impending safe mode for cloud connection without running application firmware.



***Looking for Internet***

- **Blinking Green:** The device is attempting to establish a cellular connection.





### ***Red flash SOS***

Is your device blinking red? Oh no!

A pattern of more than 10 red blinks is caused by the firmware crashing.

[support.md](#)

### ***No status LED***

If you power up your device and the status LED never comes on

## Other app information

### *Connectivity and Reporting*

- **Regular Updates:** The trap pings several times an hour to report its status and confirm network connection.
- **Disconnection Alert:** Users are alerted if the trap loses network connectivity for over an hour.

### *User Interface*

- **Indicator:** The trap's status is communicated through its armed/unarmed state indicator.
- **Command Listening:** In the unarmed state, the trap awaits user commands for further operations.

This concise guide helps users understand the operational modes of the OcuTrap, ensuring efficient and safe usage.

### *Refresh*

The refresh button will send a request to the trap for the most recent data if the trap is connected.

### *Control Button*

The control button opens a popup. You can see:

- Signal Strength
- Temperature
- Last Heard time
- Flash LED light
- Sound the alarm
- Share the trap with other users

**Expand** 

The options icon is located at the bottom of the trap tab. You can view the camera and map.

**Sharing**

You can share traps with other users within the traps options popup. Enter the email of an existing OcuTrap users email to share the trap. You cannot share with yourself, non-OcuTrap users, or double share traps. If the email fits these conditions, an email will be sent and the trap will appear at the bottom of the shared user's traps page under the Shared section.

When the trap is shared, the shared user can perform the activities of a regular trap, except the shared user cannot share the shared trap with other users.

**Changing trap name**

Click on the current name of the trap such as "52353454" or "Farm Trap". A popup will appear and you can rename the trap.

**Tips and Tricks**

Get the most out of your OcuTrap with these best practices and pro tips.

---

**Powering Off the Trap**

When you're not using the trap, **power it down properly**:

1. Hold the **power button for 3 seconds** until the device powers off
2. The trap will send a final status update before shutting down
3. This ensures a clean disconnection and protects the electronics

Proper shutdown prevents unnecessary battery drain and extends the trap's lifespan.

---

## Maximizing Battery Life

### Deployment Tips

- **Strong cellular signal** — Poor signal causes the trap to work harder to stay connected, draining battery faster
- **GPS interval** — Keep at 8 hours (default) or disable if you don't need location tracking
- **Camera timelapse** — Set to 6+ hours or disable if you only need capture photos
- **Firmware updates** — Keep updated for the latest battery optimizations

### What Drains Battery Fastest

1. Poor cellular coverage (constant reconnection attempts)
  2. Frequent GPS updates
  3. Short camera timelapse intervals
  4. Cold temperatures (reduces battery capacity)
- 

## Optimal Trap Placement

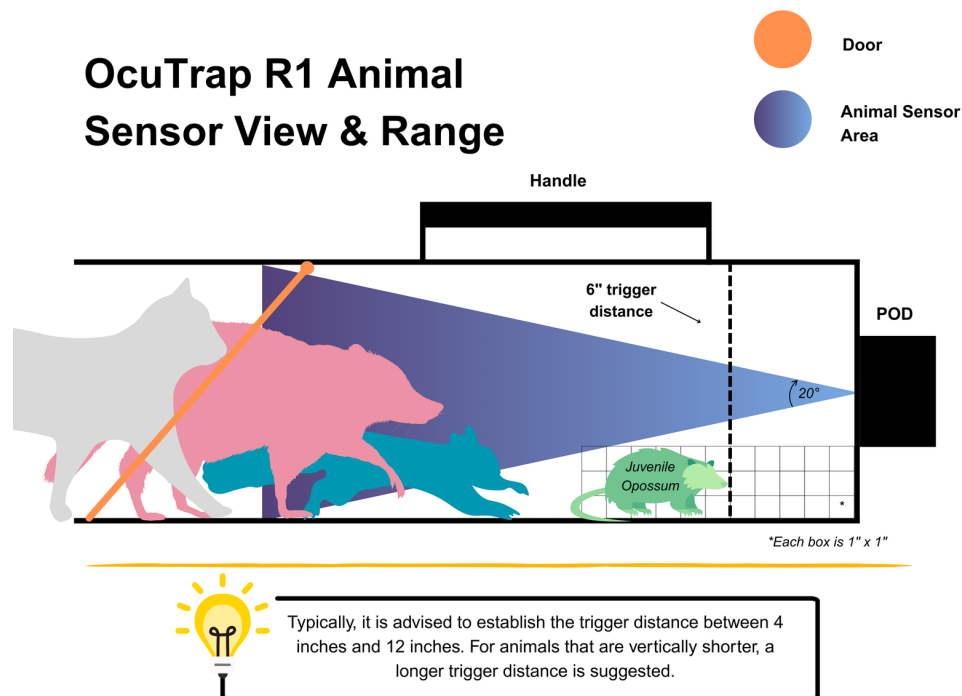
### For Best Captures

- Position trigger sensor **6-10 inches inside the cage** to ensure the animal is fully inside before the door closes
- Place bait **behind the sensor** so animals must pass through the detection zone
- Level ground helps prevent tilt alerts when armed

### For Best Connectivity

- Avoid metal buildings or dense structures that block cellular signal
  - Test signal strength before leaving the trap — check for cyan breathing LED
  - Clear sky view improves GPS accuracy
-

## Animal Sensor



## How Detection Works

### Sensor Field of View

The Time-of-Flight sensor projects a **20-degree field of view** from the center of the pod into the trap, constantly monitoring for movement.

### Detection Process

1. **Object enters detection zone** (300-450mm) — Sensor starts tracking
2. **Verification** — System confirms 3+ consecutive readings to avoid false triggers
3. **Object enters capture zone** (0-250mm) — Trigger confirmed
4. **Door closes** — Capture photo taken and alert sent

### Why This Matters

Unlike traditional traps that use a mechanical trip pan, OcuTrap uses a **distance sensor** to detect the animal's position. This method:

- Improves accuracy

- Minimizes false triggers from rain, debris, or vibration
  - Allows remote monitoring without physical trigger mechanisms
- 

## Testing the Trap

This video shows how your OcuTrap works once it's set up and ready to catch animals.

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

## Test Before Deploying

1. **Arm the trap** using the app
  2. **Wave your hand** through the detection zone
  3. **Verify** the door closes and you receive an alert
  4. **Disarm** and reset for deployment
- 

## Trigger Settings

### Capture Distance

- **Default:** 250mm (about 10 inches)
- **Adjustable range:** 125mm–1000mm
- **Tip:** Smaller values = animal must be closer before triggering

### Timing

- The sensor must register **continuous presence** before activating
  - This timing **reduces false triggers** while ensuring the animal is fully enclosed
  - Timing is optimized in firmware and cannot be manually adjusted
-

## Reducing False Triggers

If you're getting unwanted captures:

1. **Increase capture distance** — Makes the trigger less sensitive
2. **Check sensor window** — Clean any dirt, debris, or condensation
3. **Reposition the trap** — Avoid areas with blowing debris or heavy rain entry
4. **Review pre-capture photos** — See what's triggering the trap

The dual-zone verification system filters out most false triggers from rain and debris automatically.

---

## Getting Better Images

### In Daylight

- Images are automatically color
- Adjust **image rotation** if the camera view is upside down
- Use higher **camera quality** settings (3-6) for more detail

### At Night

- IR LEDs activate automatically below the dark lux threshold
- If images are too dark: Lower the **dark lux threshold** or increase **minimum IR brightness**
- If images are washed out: Decrease **maximum IR brightness**

### General Tips

- Keep the camera lens clean
  - Use **image cropping** to remove cage mesh from the frame if needed
  - Higher quality = larger files = longer transfer times
- 

## Using Pre-Capture Alerts

Enable **Pre-Capture Alerts** to get notified when an animal is approaching:

- Sends alert when object enters detection zone (before capture)
- Includes a photo of what's approaching

- 2-minute cooldown between alerts to prevent spam
- Great for monitoring activity without captures

### Use cases:

- See if non-target animals are visiting
- Monitor animal behavior patterns
- Verify trap placement is attracting targets

## Button Shortcuts

Learn the physical button controls:

| Action                | How To                                      |
|-----------------------|---------------------------------------------|
| View status           | Single press User button (5 second display) |
| Open/close door       | Double-press User button + hold 5 seconds   |
| Arm/disarm            | Press User button, then Power button        |
| Power off             | Hold Power button for 3 seconds             |
| Wake from hibernation | Press Power button                          |

## Seasonal Considerations

### Cold Weather

- Battery capacity decreases in cold temperatures
- Expect shorter runtime in winter
- Consider the 10Ah battery for extended cold-weather deployments

### Hot Weather

- Temperature alerts will notify you if internal temps exceed 45°C
- Shade the trap if possible in extreme heat
- Electronics are rated to 45°C operating temperature

### Wet Conditions

- The enclosure is weather-resistant but not waterproof
- Avoid submerging or prolonged heavy rain exposure



- Condensation on camera lens can occur — see [Condensation on the Camera](#)
- 

## Multi-Trap Management

If you're managing multiple traps:

- **Name your traps** clearly in the app for easy identification
  - **Use the map view** to see all trap locations at once
  - **Share traps** with team members using appropriate permission levels
  - **Stagger GPS intervals** if deploying many traps to spread data usage
- 

## Before You Leave the Field

Checklist before leaving your trap:

- ☐ LED shows breathing cyan (connected)
- ☐ Battery level is sufficient for deployment length
- ☐ Door opens and closes properly
- ☐ Trap is armed (yellow LED)
- ☐ GPS location is updated
- ☐ Bait is positioned behind sensor
- ☐ Trap is level and stable

## Trap Settings

### Main Settings

On the main trap settings page, you can:

- **Change the Trap Name** – Personalize each trap for easy identification.

- **Add Notes to the trap** – Add more detail to trap
- **Set Notifications** – Choose how you want to receive alerts for errors, captures, and important events.
- **Adjust Trigger Distance** – Set how far inside the trap an animal must be to trigger the door.
- **Change Image Size** – Select the photo resolution for captured images.

### Advanced / More Settings

By clicking **More Settings**, you can access advanced configuration options, including:

- **Battery Type** – Select the battery for your trap.
- **Accessory Controls** – Enable or disable accessory ports and set their timing.
- **Enhanced Door Closing** – Improve lock reliability by enabling automatic door re-locking.
- **Pre-capture Notifications** – Get alerts and images when an animal nears the trap, before capture
- **Camera Controls** – Set up time lapse, image quality, and night vision thresholds.
- **Sensor Settings** – Adjust sensor width, trigger distances, and infrared brightness.
- **Alert Thresholds** – Set up alerts for battery level, temperature, and capture intervals.

Each of these setting can be adjusted for each individual trap. All changes are applied when trap is connected. This will usually take about 1-5 minutes.

### Enhanced Door Closing

**Enhanced Door Closing** improves the reliability of the trap's door lock mechanism. When enabled, the trap will automatically cycle the door open and closed again after an initial close, ensuring the lock is fully engaged.

This feature is recommended if you want extra assurance that the trap door is securely latched, especially in environments where debris or movement might affect locking.

- **Enabled:**

After the trap door closes, it will briefly open and then re-close to re-lock, helping clear obstructions and confirm the latch is secure.

- **Disabled:**

The door will close and lock as usual, with no additional re-lock cycle.

## How to Change This Setting

To change the Enhanced Door Closing setting:

1. Click **Settings** on the trap you want to update.
2. Select **More Settings**.
3. Find **Enhanced Door Closing** and choose **Enabled** or **Disabled** from the dropdown.

**Note:** This feature is enabled by default.

## Pre-Capture Notification

### *What It Does*

**When enabled and the trap is in Armed Mode**, Pre-Capture Notification alerts you when an animal is detected approaching the trigger point. This early warning system sends a notification and a photo (if applicable), helping you stay ahead of trap activity.

## How It Works

OcuTrap uses a distance sensor to detect animals before they reach the capture zone. It monitors two key areas:

- **Early Detection Zone:** Approximately 6 inches (150mm) *before* your set capture distance

- **Example**

If your **capture distance** is set to **8 inches**, then:

- The **Primary Detection Zone** starts at **8 inches** from the sensor (this is when a capture can occur).
- The **Early Detection Zone** starts at **14 inches—6 inches before** your capture distance.

In this case, if an animal is detected at 13 inches, you'll receive an **"Early pre-capture"** alert with the message:

“Early pre-capture: 13.0 in detected”

This gives you an early warning before the trap activates at 8 inches.

If motion is detected in either zone, the trap will:

1. Take a **pre-capture photo**
2. Send a **push/email alert** with the estimated distance

### Example Alerts

- "Pre-capture: 10.0 in detected"
- "Early pre-capture: 15.7 in detected"

Units (inches or centimeters) depend on your trap's settings under **Units**.

## How to Enable or Disable

This setting is configured per trap:

1. Open the OcuTrap app or [base.ocutrap.com](https://base.ocutrap.com)

2. Tap on the trap you want to edit
3. Go to **Settings** → **More Settings**
4. Find the **Pre-Capture Notification** option
5. Toggle to **On** or **Off**

Default setting: **On**

### ***Why Use This Feature?***

- See what's approaching your trap before it's too late
- Detect non-target animals early (like pets or skunks)
- Monitor animal behavior without needing a capture
- Improve trap placement and setup based on activity patterns

### ***Need to Turn It Off?***

If you prefer to only be notified after a capture, you can disable **Pre-Capture Notification** in settings at any time. It will not affect the actual capture or release functions.

## **Maintenance**

To ensure optimal performance and longevity of your OcuTrap, please follow the maintenance checklist below. **Always power off the trap before performing any maintenance.**

## Maintenance Tasks (Every Trapping Session)

| Maintenance Task                          | Action                                                               |
|-------------------------------------------|----------------------------------------------------------------------|
| Camera Lens Cleaning                      | Wipe with a soft, lint-free cloth to remove smudges and debris.      |
| Door Operation Test                       | Verify smooth opening and closing of the door.                       |
| Trap Exterior Cleaning                    | Use a damp cloth; avoid harsh chemicals that may damage the surface. |
| Interior Components Inspection & Cleaning | Inspect the door, spring, and motor assembly for obstructions.       |
| Battery Terminal Inspection               | Look for signs of corrosion or wear; clean gently if needed.         |
| Motor and Sensor Test                     | Confirm that the motor and sensors are operating correctly.          |
| Seals and Enclosures Inspection           | Ensure that all seals and enclosures are intact and secure.          |
| Post-Adverse Weather Inspection           | Check for moisture or debris and clean as necessary.                 |

*Note: Firmware updates are automatically applied, so no manual update is required.*

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## LED Guide

### System Status Indicators

These patterns occur during power-up, connectivity, or firmware activity.

| Status               | LED Pattern        | Description                                               |
|----------------------|--------------------|-----------------------------------------------------------|
| Connected Mode       | Breathing Cyan     | Connected to the cloud and fully operational.             |
| Connecting to Cloud  | Fast Blinking Cyan | Attempting to connect to the cloud.                       |
| OTA Firmware Update  | Blinking Magenta   | Firmware update or booting in safe mode.                  |
| Looking for Internet | Blinking Green     | Searching for a cellular signal.                          |
| Red Flash SOS        | Rapid Red Blinks   | Firmware crash. Contact support if >10 blinks.            |
| No Status LED        | No Light           | Device has no power or failed to boot.                    |
| Hibernation          | LED Off            | Trap has entered low-power sleep. Wake with power button. |

Note: These modes are managed automatically. Only use the power button if the LED does not respond.

### ⚠ Low Battery Startup Behavior

When the battery is too low (below 9.6V at startup), the following will happen:

- The trap shows a **solid red LED** during boot.
- Sends a cloud error notification (if connected).
- Automatically shuts down and enters hibernation.
- Will not auto-recover — must be awakened manually using the power button.
- If voltage is still low, it will repeat the cycle.

To recover, charge or replace the battery and press the power button again.

### 🔧 User Button LED Patterns

These appear when using the physical **User Button** or **Power Button** on the device.

#### ***Device State Indicators***

| State              | LED Pattern   |
|--------------------|---------------|
| Unarmed and Open   | Solid Blue    |
| Unarmed and Closed | Solid Green   |
| Armed Mode         | Solid Yellow  |
| Armed and Captured | Solid Magenta |

#### ***Manual Door Control***

To manually open or close the trap door:

1. Press the **User Button**
2. Press again and hold for 5 seconds

| Action     | LED Pattern    |
|------------|----------------|
| Open Door  | Blinking Blue  |
| Close Door | Blinking Green |

## ***Arm/Unarm the Trap***

1. Press the **User Button**, then the **Power Button**

| Action     | LED Pattern     |
|------------|-----------------|
| Arm Trap   | Blinking Yellow |
| Unarm Trap | Blinking White  |

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## **Notes**

- LED brightness may dim in low power mode to conserve battery.
- No LED means the device is off or in hibernation.
- Any pattern not listed here may indicate a malfunction — contact support if unsure.

## **Battery Overview**

### **Current Models**

- **KBT 5000 mAh (12V)**
  - Standard battery included with OcuTrap.
  - Estimated battery life: **~21 days** per charge (under normal usage).
  - Charger: **1A (12V)**
- **KBT 10000 mAh (12V) (Coming Soon)**
  - Doubles the runtime compared to the 5000 mAh battery.
  - Charger: **2A (12V)**
  - Expected battery life: **~40+ days** per charge (under normal usage).

### **Charging**

- Use only the **provided KBT charger** to ensure safety and proper charging.
- The **5000 mAh battery** takes approximately **5–6 hours** to fully recharge.
- The **10000 mAh battery** will charge in approximately **5–6 hours** as well, thanks to its higher 2A charger.



- Always disconnect the charger once fully charged to preserve long-term battery health.

### Extending Battery Life

We are actively working on **software improvements** to increase efficiency and extend battery life. In addition, you can maximize runtime by:

- Ensuring traps are placed in areas with **strong cellular** (poor signal increases power usage).
- Keeping firmware up to date (updates include **battery optimization improvements**).
- Using the appropriate battery size for your deployment length (e.g., **10k mAh for longer-term setups**).
- Powering down the trap when not in use.

### Battery Care & Safety

- Store batteries in a **cool, dry place** when not in use.
- Avoid exposing batteries to extreme heat or moisture.
- Only use **OcuTrap-approved chargers**.
- Do not attempt to open, puncture, or modify the battery

# FAQs

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## Sharing Traps

OcuTrap allows users to share traps with others, enabling collaboration and remote monitoring. To share a trap, follow these steps:

### Requirements

- The user sharing the trap **must have an OcuTrap account**.
- The recipient must also have an OcuTrap account with a valid email address.

### How to Share a Trap

1. **Log into your OcuTrap account.**
2. Navigate to the **Sharing** section.
3. Select the **trap** you want to share.
4. Click the **plus (+) icon** to add a new user.
5. Enter the recipient's **valid email address**.
6. Confirm the sharing action.

Once shared, the recipient will have access based on their **user level**. To learn more about different access levels, see the [User Levels](#) page.

### User Levels

**Account Owners** have privileges over the entire account. Only the account owner can update billing.

**Managers** can view, share, and edit trap settings & alerts in the groups that you assign them to.

## Common questions

Can OcuTrap differentiate species like raccoons, skunks, or opossums?

Currently, OcuTrap traps any animal that enters the trap. However, you have full remote control to **release non-target animals** safely without needing to visit the trap in person.

This remote release feature allows you to:

- Minimize stress and harm to unintended animals.
- Save time and reduce unnecessary trap visits.
- Improve efficiency by focusing on target species like raccoons.

We're actively developing smarter species identification to allow selective trapping in future updates.

What is the size of the OcuTrap

10" wide, 12" tall, 34" long

How to change personal details?

Personal details would include your first name, last name, email. These can be changed when you are logged in by going to **account->edit details**. Once you edit the detail and click save your account details will be updated.

How to share trap to other accounts?

You can share your trap with other users to allow them to monitor and control it. To share a trap:

1. Open the **OcuTrap Mobile App** and select the trap you want to share.
2. Navigate to **Settings** → **Sharing** (or **Manage Users**).
3. Tap **Add User** and enter the email address of the person you want to share with.
4. Select the **permission level** for the new user (see User Levels for details).
5. The invited user will receive a notification and can access the trap once they accept.

For more details on permission levels, see [User Levels](#).

How does OcuTrap differentiate between targeted and non-targeted animals?

OcuTrap utilizes advanced sensor technology that can distinguish between different animal species based on size, shape, and other unique characteristics. This ensures that only the intended species are captured, preventing unintended captures.

Is OcuTrap humane and safe for animals?

Yes, OcuTrap is designed with animal welfare in mind. It employs a non-lethal, humane trapping mechanism that safely contains the animal without causing harm or stress.

Can I control OcuTrap remotely?

Absolutely. OcuTrap is equipped with smart technology that allows you to open and close the trap door remotely via a mobile app, giving you control from anywhere.

How do I receive notifications from OcuTrap?

Once an animal is trapped, OcuTrap sends an instant notification to your connected mobile device. You can customize notification settings in the OcuTrap app.

What maintenance does OcuTrap require?

OcuTrap requires minimal maintenance. Regularly check and clean the trap, ensure the sensors are unobstructed, and replace batteries as needed.

Is OcuTrap weather-resistant?

Yes, OcuTrap is designed to be durable and weather-resistant, making it suitable for various outdoor environments.

Can OcuTrap be used for both domestic and wild animals?

OcuTrap is versatile and can be configured to trap both domestic and wild animals, depending on the user's requirements.

What is the range of the remote control feature?

The remote control feature of OcuTrap works over any distance as long as you have an internet connection on your mobile device to access the app.

How does OcuTrap ensure the safety of non-target animals?

OcuTrap's selective targeting system and humane trapping mechanism ensure that non-target animals, if accidentally captured, are not harmed and can be safely released.

Where can I purchase OcuTrap?

OcuTrap is available for purchase through our official website and select authorized dealers. Please visit our website for more information on where to buy.

What types of animals can OcuTrap effectively trap?

OcuTrap is versatile and can be configured to trap a wide range of animals, from smaller pests like rodents to larger animals like raccoons, depending on the specific model and settings.

How long does the battery last in an OcuTrap device?

The battery life of OcuTrap depends on usage frequency, but on average, it can last several weeks to months. We recommend regular checks to ensure uninterrupted service.

Is the OcuTrap app compatible with both iOS and Android devices?

Yes, the OcuTrap app is designed to be compatible with both iOS and Android platforms, providing a wide range of accessibility for users.

Can OcuTrap be used in areas with no Wi-Fi?

Yes. OcuTrap connects exclusively via 4G LTE cellular — no Wi-Fi is required or supported. As long as there is cellular coverage in the deployment area, OcuTrap will work normally.

Does OcuTrap require a subscription service for its app or notifications?

Currently, OcuTrap does not require a subscription for basic functionality. However, additional features and services may be available with a subscription plan in the future.

Are there any ongoing costs associated with using OcuTrap?

Apart from the initial purchase, the main ongoing costs would be for battery replacement and any optional subscription services for enhanced features.

How environmentally friendly is OcuTrap?

OcuTrap is designed with environmental sustainability in mind, using eco-friendly materials and ensuring humane treatment of animals.

Can OcuTrap be used in residential areas?

Yes, OcuTrap is suitable for use in residential areas, and its humane and discreet operation makes it an ideal solution for home pest control.

What safety features does OcuTrap have?

OcuTrap includes various safety features such as secure containment of trapped animals, remote operation to minimize human-animal contact, and selective targeting to prevent capturing non-target species.

How is OcuTrap powered?

OcuTrap is typically powered by batteries, ensuring it can be deployed in a variety of locations without the need for external power sources.

Can OcuTrap withstand extreme weather conditions?

OcuTrap is built to withstand various weather conditions, including rain and extreme temperatures, ensuring reliable operation in different environmental settings.

Is there a warranty on OcuTrap devices?

Yes, OcuTrap comes with a limited warranty. The details of the warranty period and coverage can be found in the product documentation.

How user-friendly is the setup process for OcuTrap?

The setup process for OcuTrap is designed to be straightforward and user-friendly, with step-by-step instructions provided in the manual and online resources.

Can OcuTrap be linked to multiple devices?

Yes, OcuTrap allows linkage to multiple devices, enabling more than one user to monitor and control the traps.

How does OcuTrap contribute to humane animal control?

OcuTrap's design prioritizes humane capture, ensuring animals are not harmed or unduly stressed, aligning with modern animal welfare standards.

What training or knowledge is required to operate OcuTrap?

Operating OcuTrap requires minimal training. The user manual and online tutorials provide all necessary information for effective use.

Can the OcuTrap system be integrated with other smart home devices?

Currently, OcuTrap operates as a standalone system, but future developments may include integration capabilities with other smart home devices.

What happens if an OcuTrap device malfunctions?

In the unlikely event of a malfunction, users are advised to contact customer support for assistance and troubleshooting. Warranty coverage may apply for any manufacturing defects.

Is there a maximum range for the effective use of OcuTrap?

While there is no maximum range for remote monitoring as long as there is internet connectivity, the physical effectiveness of the trap will depend on appropriate placement relative to animal activity.

How does OcuTrap handle false triggers or accidental captures?

OcuTrap's advanced sensors are designed to minimize false triggers. In case of an accidental capture or false trigger:

- **Remote Release:** You can release the animal remotely using the app without visiting the trap in person.
- **Notification Review:** Check the captured images to verify whether an animal was caught before taking action.
- **Sensitivity Adjustment:** Adjust the sensor sensitivity in trap settings if you experience frequent false triggers.
- **Pre-Capture Notifications:** Enable pre-capture notifications to review images before the door closes automatically.

These features help ensure efficient operation while minimizing disruptions from false triggers.

Can the OcuTrap be used in commercial settings like farms or warehouses?

Yes, OcuTrap is designed for versatility and can be effectively used in both commercial settings like farms and warehouses, and residential areas.

How does OcuTrap ensure the humane treatment of trapped animals?

OcuTrap uses a non-lethal, containment approach that minimizes stress and harm to the trapped animals, adhering to humane animal treatment standards.

Is there a limit to the number of OcuTrap devices that can be managed through the app?

The app is designed to manage multiple devices efficiently, though the exact number may depend on the specific app version or subscription plan.

Can OcuTrap be used for specific types of pests like rodents or raccoons?

Yes, OcuTrap can be configured to target specific types of pests, including rodents, raccoons, and other animals, depending on the user's needs.

Are there different sizes or models of OcuTrap available?

OcuTrap comes in various sizes and models to accommodate different types of animals and use cases.

How does OcuTrap deal with multiple animals approaching the trap?

OcuTrap's smart technology can detect and manage multiple animal presences, ensuring effective and selective trapping.

Is there a mobile app update required to keep using OcuTrap?

Regular updates to the mobile app are recommended to ensure optimal performance and access to the latest features.

How secure is the data and information in the OcuTrap app?

OcuTrap employs robust security measures to protect user data and information within the app.



Can OcuTrap be used in sensitive environments like schools or hospitals?

Yes, OcuTrap's humane and discreet operation makes it suitable for use in sensitive environments such as schools and hospitals.

What type of customer support does OcuTrap offer?

OcuTrap provides comprehensive customer support, including a helpdesk, email support, and online resources.

How does OcuTrap contribute to sustainable pest control practices?

OcuTrap's design and operation align with sustainable pest control practices, minimizing environmental impact and promoting humane treatment.

Can OcuTrap's settings be customized according to specific requirements?

Yes, users can customize OcuTrap's settings to meet specific trapping requirements and conditions.

Is there a trial period or demo available for OcuTrap?

Potential customers can contact OcuTrap for information about any available trial periods or demos.

How often do I need to check the OcuTrap physically?

Physical checks are minimized due to remote monitoring, but periodic checks are recommended for maintenance.

Does OcuTrap work in areas with high animal populations?

OcuTrap is effective even in areas with high animal populations, offering efficient and selective trapping.

What are the power options for OcuTrap (battery, solar, etc.)?

OcuTrap primarily uses batteries, but there may be other power options like solar, depending on the model.

Can OcuTrap be used in conjunction with other pest control methods?

Yes, OcuTrap can be integrated into a broader pest control strategy, complementing other methods.

How resistant is OcuTrap to tampering by animals or humans?

OcuTrap is designed to be tamper-resistant, ensuring its effectiveness and durability against animal and human interference.

What should I do if an animal is injured in the OcuTrap?

In the unlikely event of an injury, contact local animal control or a veterinarian immediately. OcuTrap is designed to minimize such occurrences.

Can OcuTrap be used in all weather conditions?

OcuTrap is built to operate in a range of weather conditions, ensuring reliable performance year-round.

Are there any additional accessories required for OcuTrap?

OcuTrap comes with all necessary components, but additional accessories may be available to enhance its functionality.

How do I know if OcuTrap is the right solution for my pest control needs?

Consider your specific pest control requirements and consult with an OcuTrap expert to determine if it's the right solution for you.

Does OcuTrap have an impact on local wildlife populations?

OcuTrap's selective and humane approach minimizes negative impacts on local wildlife populations.

How quickly can OcuTrap be deployed in an area?

OcuTrap is designed for easy and quick deployment, allowing for rapid setup in any chosen area.

Is OcuTrap effective in urban environments?

Yes, OcuTrap is highly effective in urban settings, addressing common urban wildlife management challenges.

Can OcuTrap be used in rural or wilderness areas?

OcuTrap is also suitable for rural or wilderness areas, provided there is connectivity for remote monitoring.

Are there any environmental considerations to be aware of when using OcuTrap?

OcuTrap is environmentally friendly, but it's always important to consider local regulations and ecological impacts.

Can OcuTrap be used for scientific or research purposes?

Yes, OcuTrap can be an effective tool for wildlife research and scientific studies, given its precision and data capabilities.

Is there a community forum or user group for OcuTrap?

Users can check the OcuTrap website for information on community forums or user groups for sharing experiences and tips.

How does OcuTrap compare with traditional trapping methods?

OcuTrap offers significant advantages over traditional methods, including humane treatment, remote monitoring, and efficiency.

Are there any ongoing software updates for OcuTrap?

OcuTrap receives regular software updates to enhance its functionality and user experience.

How do I dispose of or recycle an old OcuTrap unit?

Contact OcuTrap customer service for guidance on environmentally responsible disposal or recycling of old units.

Can I rent or lease OcuTrap instead of purchasing it outright?

Information about rental or leasing options can be obtained by contacting OcuTrap directly.

How can I provide feedback or suggestions for OcuTrap?

OcuTrap welcomes user feedback and suggestions, which can be submitted through the app or the website.

Is training provided for large-scale deployments of OcuTrap?

For large-scale deployments, OcuTrap can provide training and support to ensure effective use of the technology.

Can OcuTrap be programmed to release trapped animals remotely?

The current version of OcuTrap requires manual release of animals, but future versions may include remote release features.

Are there any known issues or limitations with OcuTrap?

Like any technology, OcuTrap may have limitations depending on the specific environment or use case. Consult the product documentation for details.

What are the dimensions and weight of an OcuTrap unit?

The dimensions and weight vary by model. Specific details can be found in the product specifications.

Can OcuTrap be used to trap invasive species?

OcuTrap can be an effective tool for managing invasive species, depending on local regulations and the specific species.

Is OcuTrap suitable for trapping in marine or coastal environments?

While OcuTrap is weather-resistant, its suitability for marine or coastal environments depends on the specific model and environmental conditions.

How does OcuTrap handle power outages or connectivity issues?

OcuTrap is designed to retain basic functionality during power outages, but remote features may be limited without connectivity.

Can OcuTrap be used for trapping birds or flying animals?

OcuTrap is primarily designed for ground-based animals. Its effectiveness for birds or flying animals may be limited.

What measures are in place to prevent OcuTrap from being stolen or vandalized?

OcuTrap includes security features to deter theft or vandalism, but users should also take appropriate precautions based on the deployment location.

How does OcuTrap integrate with other smart technology or IoT devices?

While OcuTrap operates as a standalone device, future iterations may offer integration with other smart technology or IoT devices.

What is one-player id?

This ID is used to maintain push notifications to your mobile device. If the field is blank under your account page, you will not receive push notifications. Please allow notifications on your mobile device to enable the capture of the one-player ID. If the issue persists, please delete the OcuTrap app and redownload it.

## Safe-mode

**Prerequisite:** Make sure device has enough battery charge. If not turning on, charge the battery and try again.

**To put your device in Safe Mode:**

1. Hold down BOTH buttons (RESET and MODE).
2. Release only the RESET button, while holding down the MODE button.
3. Wait for the LED to start blinking magenta
4. Release the MODE button.

Before entering safe mode, the device will proceed through the normal steps of connecting to the cloud; blinking green, blinking cyan, and fast blinking cyan. If you're unable to connect to the cloud, you won't actually end up with breathing magenta, but execution of application firmware will still be blocked - so you are in a "sort-of safe mode".

## Battery

### Features

**High-Quality Batteries:** Each pack is built with premium batteries to provide powerful energy.

**Built-in Protection Board:** Prevents over-charge, over-discharge, over-voltage, and short

### ***Charging Process:***

#### **1. Identifying the Connectors:**

- Your battery pack comes with **two connectors**:
  - The **yellow connector** is for connecting to the OcuTrap device.
  - The **black connector** is used exclusively for charging the battery.

#### **2. Charging the Battery:**

- Plug the **black connector** into the supplied **12V battery charger**.
- Connect the charger to a power source.
- The charger's indicator light will show:
  - **Red light**: Battery is charging.
  - **Green light**: Battery is fully charged.

#### **3. Charging Duration:**

- If the battery is **completely drained**, it will take approximately **4-6 hours** to fully charge.

#### **4. Disconnecting:**

- Once the charger's light turns **green**, the battery is fully charged and can be disconnected from the charger.

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### ***Package Includes:***

- **1 x 12V 5200mAh battery pack**
- **1 x 12V battery charger**

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### ***Specifications:***

- **Battery Pack Size**: 70 x 55 x 40 mm (L x W x T)
- **Battery Weight**: Approx. 295g
- **Charger Cable Length**: 90 cm

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Always ensure to charge your battery properly to maintain optimal performance and battery life. If you encounter any issues, feel free to contact our support team.

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Let me know if you need any more changes or additions!

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### ***Package Includes***

- 1 x 12V 5200mAh battery pack
- 1 x 12V battery charger

### ***Specifications***

- **Battery Pack Size:** 70 x 55 x 40 mm (L x W x T)
- **Battery Weight:** Approximately 295g
- **Charger Cable Length:** 90 cm

## **Updating Firmware**

OcuTrap firmware updates are delivered automatically over-the-air (OTA) to ensure your device has the latest features and security improvements.

### **How Firmware Updates Work**

#### **Automatic Updates**

- Firmware updates are pushed automatically when your OcuTrap is connected to the internet.
- Updates typically occur during low-activity periods to minimize disruption.
- Your trap will briefly restart after an update is applied.

#### **Update Requirements**

For a successful firmware update:

- The trap must be powered on and connected to cellular.
- Battery level should be above 20% to prevent interruption during the update.
- The trap should remain powered during the update process (usually 5-15 minutes).

## Checking Your Firmware Version

To view your current firmware version:

1. Open the **OcuTrap Mobile App**.
2. Select your trap from the device list.
3. Navigate to **Settings** or **Device Info**.
4. The firmware version is displayed under device details.

## Troubleshooting Update Issues

### Update Not Installing

- Ensure the trap has a stable internet connection.
- Verify the battery level is sufficient.
- Try power cycling the trap by removing and reinserting the battery.

### Trap Unresponsive After Update

- Wait 5 minutes for the trap to complete its restart sequence.
- If the trap remains unresponsive, perform a manual power cycle.
- Contact [support](#) if issues persist.

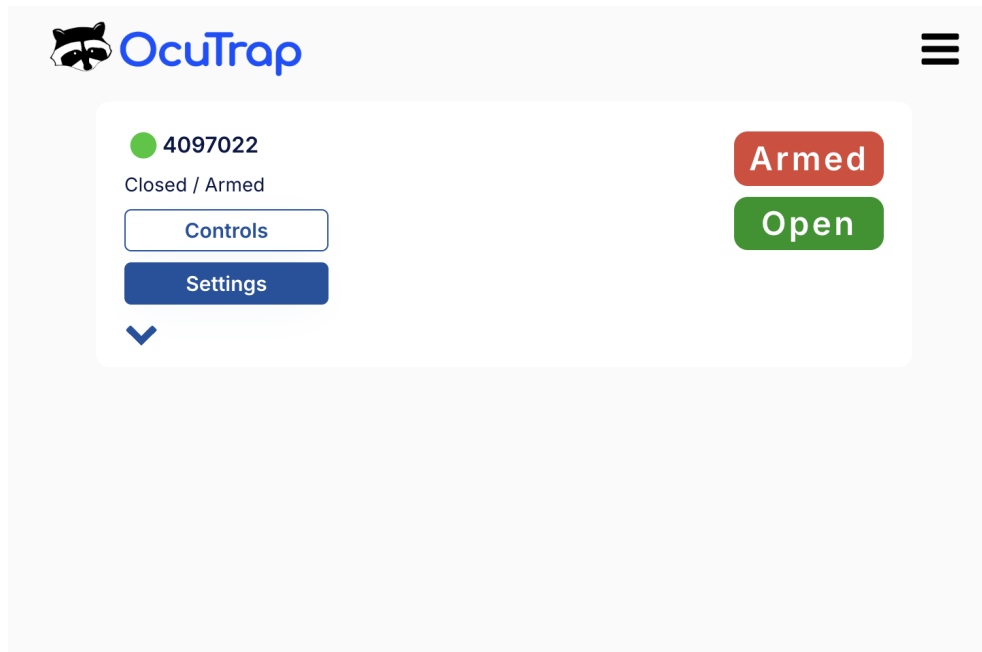
## Release Notes

For information about the latest firmware features and fixes, check the **Updates** page on [docs.ocutrap.com](https://docs.ocutrap.com) or contact support.

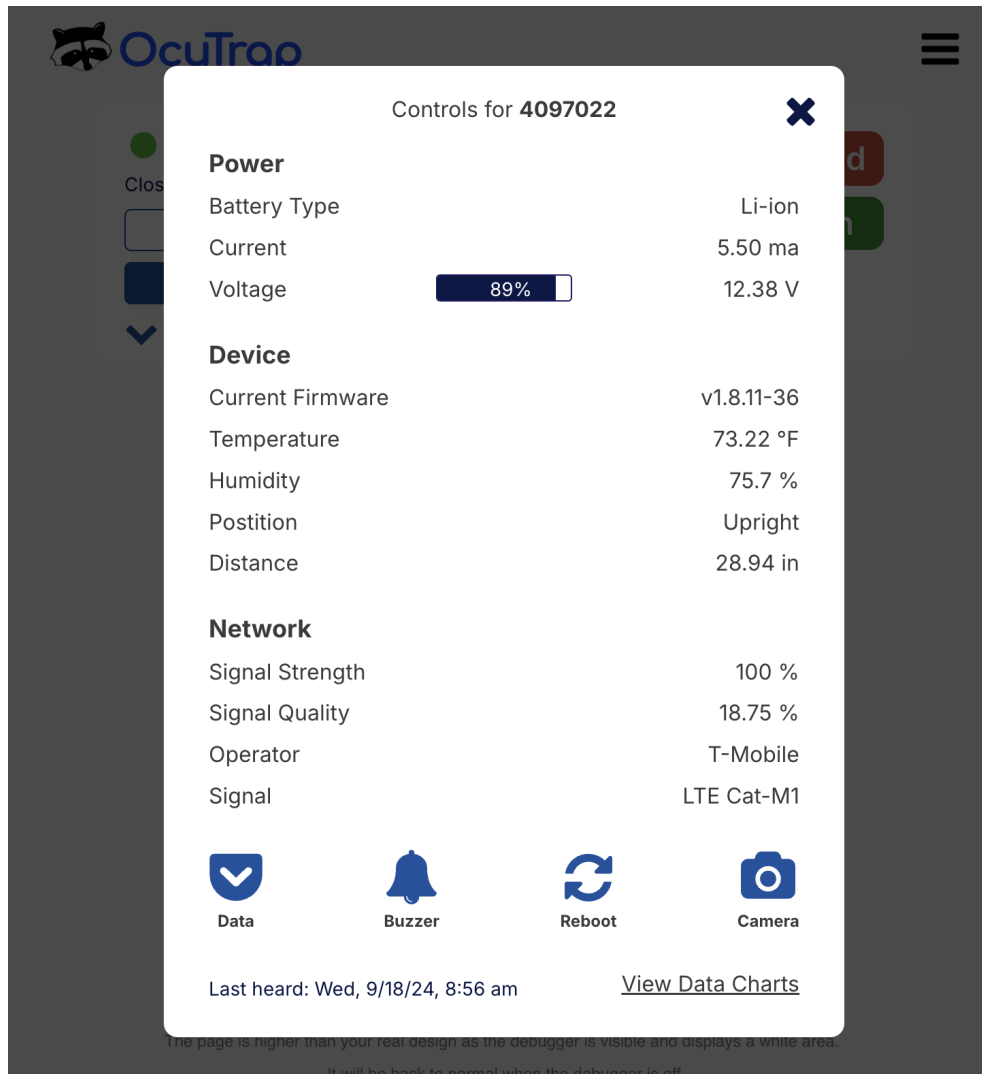
## Manually taking an image

Go to your console page. Click on the controls icon





Click on the camera icon on the bottom right

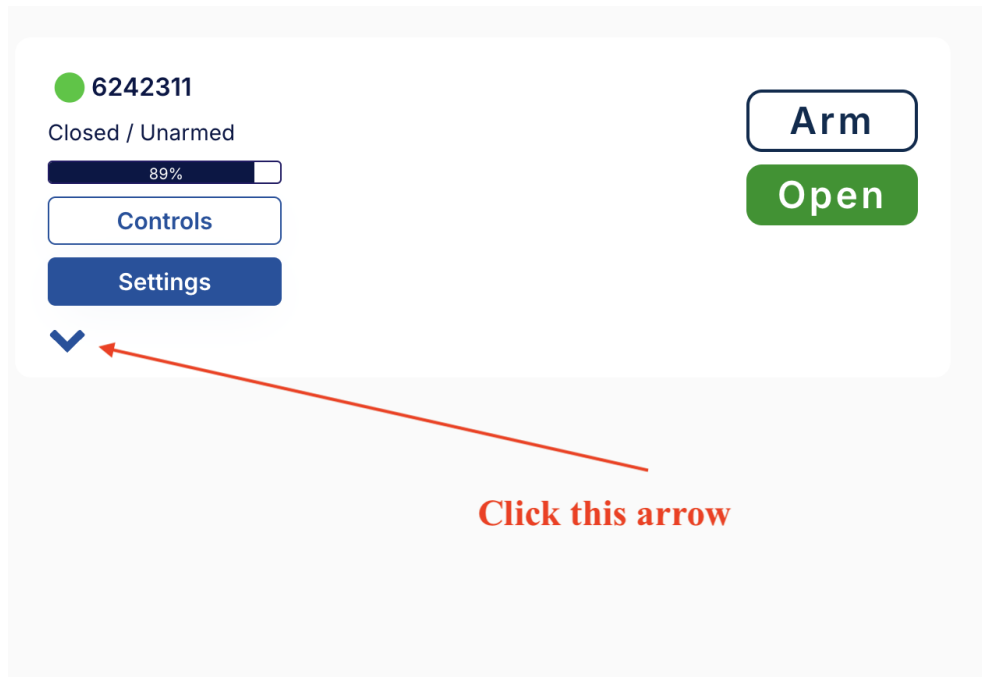


### Fast image request

Tap the lightning bolt icon to request an image faster. The trap returns a lower-quality image in exchange for the speed boost — useful when you need to check on something right now.

### Seeing Camera view

To open the camera view, click the downward arrow icon located beneath the settings button.



## Taking higher quality images

Manage: 4097022

[Change Trap Name](#)

**Notifications**

Error Push-notification

Alert Push-notification

Capture Push-notification

**1 Larger is better quality but takes longer to send**

Image Size

Trigger distance

☐ none

Monitoring Mode

☒ Enabled

☐ Enabled

[More Settings](#)

**2**

[Update now](#)

**Trap Order**

Position: 1

SN: 0a10aced202194944a051c4c

## GPS

OcuTrap uses an integrated u-blox GPS module to provide location tracking and mapping capabilities. This guide explains the GPS settings and functionality to help you get the most accurate location data for your traps.

## How GPS Works on OcuTrap

GPS is **battery-optimized by design**. Rather than continuously tracking location, OcuTrap uses strategic update intervals to maximize battery life while keeping you informed of trap locations.

### *Default Behavior*

- **Update Interval:** Every 8 hours (default)
  - **First Boot Delay:** 5-minute delay after power-on before first GPS acquisition
  - **Automatic Capture Updates:** GPS automatically triggered when a capture occurs
  - **Fix Requirements:** Minimum 5 satellites with a 3D fix for valid position
- 

## GPS Settings

### *GPS Interval*

- Controls how frequently the GPS updates location data
- **Default:** 8 hours (recommended for battery life)
- Can be disabled to conserve battery life
- Located in Settings → GPS Interval
- Manual updates still possible through the interface when disabled

### *GPS Status Indicators*

When viewing trap locations, you'll see key metrics:

- **Satellites Connected:** Number of GPS satellites currently in use (e.g., "8 connected"). More satellites = better accuracy.
- **Last Updated:** How long ago the GPS position was updated (e.g., "6 Hours Ago").
- **Radius:** The trap's last location is within the approximate radius.

## Best Practices

### ***Optimal GPS Performance***

- Place the OcuTrap outdoors with clear sky view for best results
- **First fix:** Allow up to 3 minutes for initial GPS acquisition after power-on
- **Subsequent fixes:** Typically acquired within 2 minutes
- System will timeout if no fix is acquired within the timeout period
- More satellites generally means better accuracy
- Buildings, dense foliage, and urban canyons can reduce accuracy

### ***Manual Updates***

- In controls, click on data button to request a GPS update
- Only works when GPS is not disabled in settings
- Useful for verifying position without waiting for next interval

### ***Access Levels***

- GPS viewing capabilities are restricted by user access level
- Owner and Manager level users and above can view all trap locations in map tab
- Lower access levels may have restricted viewing capabilities

## Troubleshooting

### ***No Fix Available***

1. Ensure device is outdoors with clear view of sky
2. Wait up to 2 minutes for initial fix
3. Check GPS Interval setting is not disabled
4. Try manual update by clicking location data
5. If problems persist, verify no physical obstructions are blocking GPS antenna

### ***Poor Accuracy***

- Move device to location with clearer sky view
- Wait for more satellites to be acquired
- Verify Fix type is 3 for best accuracy

- Consider environmental factors (buildings, trees, etc.)

### Battery Considerations

- GPS usage impacts battery life significantly
- **Default 8-hour interval** is optimized for multi-week deployments
- Shorter intervals provide more frequent updates but reduce battery life
- Disable GPS when location tracking not needed
- GPS is automatically disabled in Low Power mode to conserve battery
- Poor cellular signal in combination with GPS can increase power consumption

### Map Interface

- Toggle between Map and Satellite views
- Terrain overlay available for topographical reference
- Zoom controls for detailed area inspection

## Weather & Environmental Guidelines

### General Weatherproof Design

OcuTrap's exterior is designed to resist harsh weather, including rain, wind, and moderate environmental challenges. However, please keep in mind:

- **Internal Electronics:** Although the enclosure is weatherproof, internal electronics must not be submerged in water.
- **Durability:** The trap is built to endure outdoor conditions, but taking precautions in extreme weather can prevent potential damage.

### Operating & Charging Temperature Guidelines

#### *Operating Temperature*

For best performance:

- **Ideal Range:** Operate OcuTrap between **0°C (32°F)** and **40°C (104°F)**.

## ***Charging Temperature***

Since OcuTrap is powered by a lithium-ion battery:

- **Recommended Charging Range:** Charge the battery in ambient temperatures between **0°C (32°F)** and **45°C (113°F)**.
- **Performance Consideration:** Charging outside of this range can affect battery efficiency and lifespan.

## ***Temperature Monitoring***

The OcuTrap R1 model features an integrated temperature sensor that monitors the internal environment. This sensor alerts users if the internal temperature reaches adverse levels, helping to prevent potential issues.

## **Handling Extreme Weather Conditions**

### ***Extreme Heat and Cold***

While OcuTrap is capable of operating in a broad temperature range, long-term exposure to temperatures outside the ideal operating conditions can:

- Reduce overall performance.
- Diminish battery life.

### ***Icing Conditions***

In freezing weather:

- The moving door mechanism (which opens and closes) may freeze shut.
- To maintain functionality, avoid operating OcuTrap in conditions where icing is likely.

For comprehensive cold weather guidance including battery care, condensation prevention, and seasonal deployment tips, see [Cold Weather](#).



## ***Heavy Rain and Flooding***

Even though the external design is weatherproof:

- Ensure that the internal electronics are not exposed to submersion.
- During heavy rain or flooding, consider relocating or protecting the trap to keep the internal components dry.

## **Battery-Specific Guidance**

To maximize the performance and longevity of the lithium-ion battery:

- Adhere to the recommended operating and charging temperature guidelines.
- Avoid exposing the battery to prolonged extreme temperatures.
- No additional maintenance is required beyond following these guidelines.

For more battery information, see [Battery Overview](#).

## **Related Pages**

- [Cold Weather](#) — Detailed guide for winter deployments
- [Battery Overview](#) — Complete battery information
- [Condensation on the Camera](#) — Troubleshooting camera condensation

## **Cold Weather**

Operating OcuTrap in cold weather requires some additional considerations.

This guide consolidates all cold weather-related information to help you maintain optimal performance during winter months.

## **Temperature Ranges**

### **Operating Temperature**

- **Ideal Range:** 0°C (32°F) to 40°C (104°F)
- **Extended Range:** The trap can function in temperatures as low as 0°C (32°F). However, prolonged exposure is not recommended as it may impact performance and battery longevity. Possible issues include door and motor sticking, icing, and motor seizing.

## Charging Temperature

- **Recommended Range:** 0°C (32°F) to 45°C (113°F)
- **Important:** Charging outside this range can affect battery efficiency and lifespan. Bring the battery indoors to charge if ambient temperatures are below freezing.

## Temperature Alerts

The OcuTrap R1 model features an integrated temperature sensor that monitors internal conditions:

- **Low Temperature Alert:** Triggers at -10°C (14°F)
- **Alert Interval:** Every 8 hours (configurable from 0–48 hours)

## Battery Performance in Cold Weather

Cold temperatures significantly affect lithium-ion battery performance:

- **Reduced Capacity:** Battery capacity decreases in cold temperatures, sometimes dramatically
- **Shorter Runtime:** Expect shorter runtime during winter deployments
- **Recommended Solution:** Consider the 10Ah battery (standard for US shipments) for extended cold-weather deployments to compensate for reduced capacity

## Battery Care Tips

- Store batteries in a cool, dry place when not in use
- Avoid leaving batteries in freezing conditions for extended periods
- If possible, swap batteries more frequently during cold weather operations
- Charge batteries at room temperature for best results

## Ice and Freezing Conditions

### Door Mechanism

In freezing weather, the door mechanism may be affected:

- **Frozen Door:** The moving door mechanism (which opens and closes) may freeze shut

- **Motor Seizing:** In icing conditions, the door motor may seize if the arm becomes frozen, preventing the motor from moving
- **Recommendation:** Avoid operating OcuTrap in conditions where icing is likely

### Preventing Ice-Related Issues

- Check weather forecasts before deployment
- If ice formation is expected, consider temporarily retrieving the trap
- After ice events, inspect the door mechanism before resuming operation
- Allow frozen components to thaw naturally—do not force the door open

### Condensation

Temperature swings common in cold weather (such as morning/evening transitions) can cause condensation issues:

#### Symptoms

- Foggy or "milky" images that typically clear after a few minutes
- This is caused by external condensation from temperature changes

#### Solutions

- **Immediate Fix:** Place a fresh silica gel pack inside the trap pod before closing
- **Deep-Dry Procedure:** Leave the POD cracked open in a warm, ventilated spot for 24 hours, or seal in a zipper bag with desiccant packs for 12–18 hours

### Seasonal Deployment Tips

#### Before Winter

- Fully charge batteries and consider having spares ready
- Inspect door mechanism and ensure it moves freely
- Add fresh silica gel packs to prevent condensation
- Test the trap to confirm all systems function properly

## During Winter

- Monitor battery levels more frequently
- Check for ice buildup after freezing rain or sleet events
- Be prepared for reduced check-in frequency if battery is depleted faster
- Review temperature alerts in the app

## End of Winter

- Perform a full inspection of the trap
- Check for any moisture intrusion or corrosion
- Replace silica gel packs
- Verify door mechanism operates smoothly after the cold season

## Related Pages

- [Weather & Environmental Guidelines](#) — Full environmental guidelines including heat and rain
- [Battery Overview](#) — Complete battery information
- [Tips and Tricks](#) — General deployment tips
- [Condensation on the Camera](#) — Detailed condensation troubleshooting

## Camera

The **OcuTrap camera** provides **daytime and night vision capabilities**, allowing users to configure image capture settings based on lighting conditions.

---

### Camera Time-Lapse Mode

OcuTrap can **automatically capture images** at set intervals while the trap is in **armed mode**.

### *Configurable Option*

- **Photo Capture Frequency** — Defines how often the camera captures an image while the trap is armed.

If images are too frequent, **reduce the frequency** to conserve battery and data. If you need more monitoring, **increase the frequency** to capture more activity.

---

## Camera Image Settings

These settings control **image quality, resolution, and night vision brightness**.

- **Image Quality** – Adjusts resolution and compression. Higher quality means clearer images but **increased data usage**.
- **Image Size** – Defines the resolution of captured images.
- **Maximum IR Brightness** – Adjusts infrared light intensity in **night vision mode** to **prevent overexposure or underexposure**.

If **daytime images look fine but nighttime images are too dark**, **increase *Max IR Brightness***.

If **nighttime images are washed out or too bright**, **lower *Max IR Brightness***.

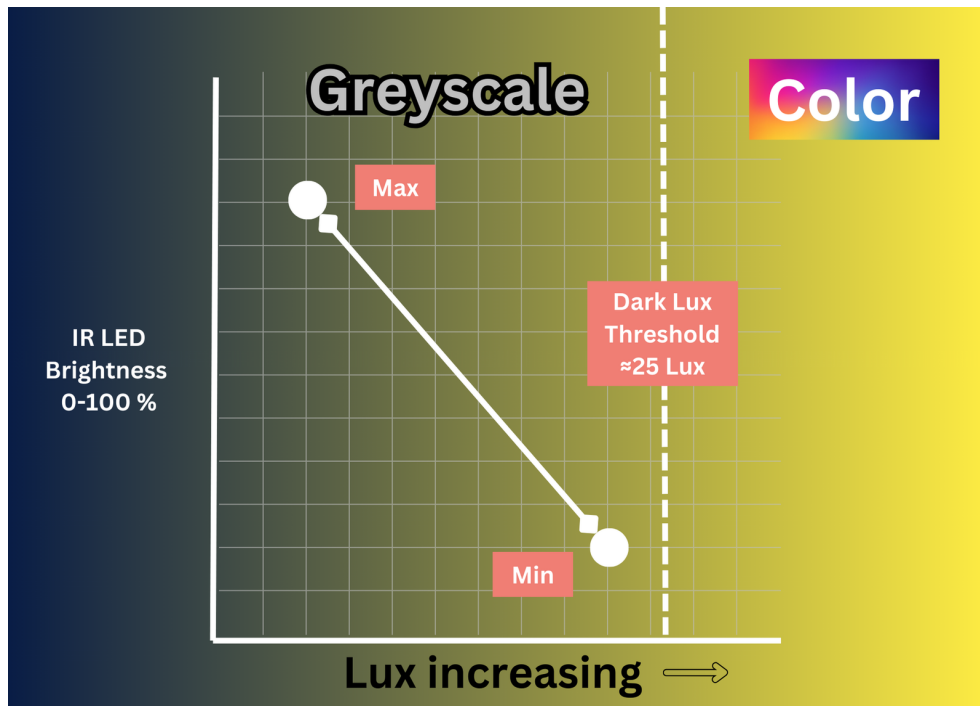
## Limits for IR Brightness

| Setting           | Min Value | Default Value | Max Value |
|-------------------|-----------|---------------|-----------|
| Max IR Brightness | 0         | 50            | 100       |
| Min IR Brightness | 0         | 20            | 100       |

---

## Night Vision & Light Adaptation

**Most users do not need to adjust light settings**, as the camera is designed to work automatically. However, users can fine-tune these settings for optimal image clarity in specific environments.



The camera **automatically adjusts** between **color mode** (daytime) and **night vision (greyscale)** based on ambient light levels.

### *Automatic Light Switching*

- **Dark Lux Threshold**
  - If ambient light **falls below this value**, the camera activates **night vision mode** (greyscale) and **enables IR**.
- **Dynamic Light Adaptation**
  - When light is **between the two thresholds**, the IR brightness **gradually adjusts** to optimize visibility.

## Limits for Light Thresholds

| Setting                   | Min Value | Default Value | Max Value |
|---------------------------|-----------|---------------|-----------|
| <b>Dark Lux Threshold</b> | 10.0      | 20.0          | 100.0     |

If **daytime images are still in greyscale**, lower *Dark Lux Threshold* so the camera switches to color mode sooner.

If **nighttime images are still in color and too dark**, increase *Dark Lux Threshold* to enable greyscale mode earlier.

## Understanding and Adjusting Image Quality

If you experience **image quality issues**, use the guide below to fine-tune your settings:

| Issue                                                | Cause                                 | Solution                                                                     |
|------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------|
| <b>Daytime images appear too dark</b>                | <i>Dark Lux Threshold</i> is too high | Lower <i>Dark Lux Threshold</i> so the camera switches to color mode sooner. |
| <b>Daytime images are blurry</b>                     | Low image quality setting             | Increase <b>Image Quality</b> in settings.                                   |
| <b>Nighttime images are too dark</b>                 | IR brightness is too low              | Increase <i>Max IR Brightness</i> to enhance night vision.                   |
| <b>Nighttime images are too bright or washed out</b> | IR brightness is too high             | Lower <i>Max IR Brightness</i> to prevent overexposure.                      |
| <b>Images have too much glare</b>                    | Reflections from IR light             | Adjust trap positioning or lower <i>Max IR Brightness</i> .                  |
| <b>Images are too pixelated</b>                      | Low resolution setting                | Increase <b>Image Size</b> or <b>Image Quality</b> .                         |

## How OcuTrap Adapts to Different Light Conditions

| Light Condition                                           | Camera Mode                     | IR Brightness                                   |
|-----------------------------------------------------------|---------------------------------|-------------------------------------------------|
| <b>Bright daylight</b> (above <i>dark lux threshold</i> ) | <b>Color</b>                    | <b>Off</b>                                      |
| <b>Low light/Night</b> (below <i>dark lux threshold</i> ) | <b>Greyscale (night vision)</b> | <b><i>Between Max and Min IR Brightness</i></b> |

## Summary of Best Practices

### For Most Users:

- No need to adjust light settings—the camera adapts automatically.

### If Adjustments Are Needed:

- **Improve night images** → Increase `maxIrBrightness`.
- **Fix overexposed IR images** → Lower `maxIrBrightness`.
- **Make daytime images clearer** → Increase **Image Quality**.

## Power Modes

OcuTrap is designed with multiple power modes to **maximize battery life** while maintaining functionality in the field. These modes ensure that the device can operate efficiently for extended periods while allowing users to interact when needed.

The power management system optimizes energy consumption by adjusting connectivity, sensor activity, and LED brightness based on usage. This allows for **extended deployment without frequent battery replacements or recharges**.

**Quick Reference:** OcuTrap has 6 power modes ranging from full operation to complete shutdown. The device automatically transitions between modes based on activity, battery level, and armed state.

---

## Power Modes

### 1. Normal Power Mode

- **Description:** This is the highest power state where OcuTrap is fully operational.
  - **When Active:** When the user is interacting with the device through commands or pressing buttons.
  - **Indicators:** LED at full brightness.
-



## 2. Low Power Idle Mode

- **Description:** A power-saving mode where the device reduces energy consumption while waiting for user interaction.
  - **When Active:** After a period of inactivity in normal power mode.
  - **Indicators:** LED brightness is dimmed.
- 

## 3. Low Power Armed Mode

- **Description:** Same as Low Power Idle, but the trap is armed and ready to capture an animal.
- **When Active:** When the trap is armed and waiting.
- **Indicators:** LED brightness is dimmed.
- **Sensor Activity:** Time-of-Flight sensor operating at ~300ms intervals (optimized for battery)

**Important:** When in **armed mode**, the device will **not enter hibernation** unless a **very low battery event** occurs.

---

## 4. Sleep Mode

- **Description:** A deeper power-saving state where most systems are powered down except essential wake functions.
  - **When Active:** Extended periods of inactivity without being armed.
  - **Indicators:** LED is powered off.
  - **Power Consumption:** Very low — most sensors disabled.
  - **Wake Events:** Button press, scheduled check-in, or incoming cloud command.
- 

## 5. Armed Sleep Offline Mode

- **Description:** A specialized armed mode that extends battery life further by periodically checking in with the cloud while keeping the internet disconnected in between. The trap remains armed and monitoring for captures even when offline.

- **When Active:** When armed, but conserving power between check-ins, or when cellular connectivity is unavailable.
  - **Check-In Interval:** Every **20 minutes** (fixed, cannot be changed by the user).
  - **Power Consumption:** Lower than Low Power Armed Mode due to reduced network usage.
  - **Indicators:** LED flashes at 3-second intervals.
  - **Capture Behavior:** If a capture occurs while offline, the trap will report it at the next check-in.
- 

## 6. Hibernation Mode

- **Description:** The lowest power state where the device is completely inactive. No communication is possible in this mode.
- **When Active:**
  - If the **battery is too low** to continue operation.
  - If the device is **idle for 1 hour and 45 minutes** and **not armed**. (*Unless prevent idle hibernation is enabled — Coming Soon*)
  - If the **power button is held down for 3+ seconds**.
- **Power Consumption:** Minimal.
- **Indicators:** LED is powered off.
- **Recovery:**
  - If due to **low battery**, the user must **replace or recharge** the battery.
  - If due to **idle timeout or power button press**, the user must **press the power button to wake the device**.

**Note:** In hibernation mode, **OcuTrap cannot receive messages or send notifications**.

---

## Power Mode Transitions

### *Automatic Transitions:*

- **Normal Power Mode** → **Low Power Idle Mode** (after inactivity).
- **Low Power Idle Mode** → **Low Power Armed Mode** (when armed).
- **Armed Sleep Offline** → **Cloud Check-in** (every 20 minutes).
- **Any Mode** → **Hibernation** (if low battery).

- **Low Power Idle Mode** → **Hibernation** (if unarmed and idle for 1 hour 45 minutes).

### ***User-Controlled Transitions:***

- **Power Button Press:** Can wake the device from hibernation.
  - **Sending a Command:** Resets the idle timer and returns to normal power mode.
- 

### **Battery & Power Alerts**

To ensure users are aware of power status, **OcuTrap sends battery warnings** at:

- **20% Battery** – Low battery warning.
- **10% Battery** – Critical battery warning.
- **Hibernation** – Final alert before shutdown.

These alerts help prevent unexpected downtime and allow users to take action before the device powers off.

---

### **Future Enhancements (*Coming Soon*)**

- **Prevent Idle Hibernation** setting to keep the device awake indefinitely when unarmed.
-

## Summary Table

| Power Mode                  | Description                                | LED Status      | Can Receive Commands?    | Can Send Data?           |
|-----------------------------|--------------------------------------------|-----------------|--------------------------|--------------------------|
| <b>Normal Power Mode</b>    | Full power, user interaction               | Full brightness | ✓ Yes                    | ✓ Yes                    |
| <b>Low Power Idle Mode</b>  | Reduced power while waiting                | Dimmed          | ✓ Yes                    | ✓ Yes                    |
| <b>Low Power Armed Mode</b> | Trap is armed, waiting                     | Dimmed          | ✓ Yes                    | ✓ Yes                    |
| <b>Sleep Mode</b>           | Deep sleep, most systems off               | Off             | ✓ Yes (wakes on command) | ✓ Yes (when awake)       |
| <b>Armed Sleep Offline</b>  | Periodic check-ins, no internet in between | Flashing (3s)   | ✗ No (Between check-ins) | ✓ Yes (During check-ins) |
| <b>Hibernation</b>          | Fully powered down, no communication       | Off             | ✗ No                     | ✗ No                     |

## Unarmed Hibernation

Applies to OcuTrap R1 firmware v1.12.3-218 and later

### Overview

The **unarmedHibernation** setting controls whether the device will enter hibernation mode after extended periods of inactivity while in the **unarmed state**. This setting helps manage battery life and power consumption when the trap is not in use.

### Location

This setting can be found in each traps **Settings** popup within the OcuTrap app.

### Setting Type

- **Type:** true / false
- **Default:** true

## Functionality

### When Enabled ( `true` )

- The device will enter **hibernation mode** after approximately **2 hours** of inactivity while unarmed.
- A **warning notification** is sent **15 minutes** before entering hibernation.
- The device **completely powers down** to conserve battery.
- To wake up, press the **power button**.

### When Disabled ( `false` )

- The device remains in **low power mode indefinitely** while unarmed.
- It **will not enter hibernation**, even after extended inactivity.
- Cloud connectivity and sensor monitoring remain **active**, but this consumes more power.

## Important Notes

- This setting **only affects the unarmed state**.
- It does **not** impact hibernation triggered by **low battery** or **manual hibernation commands**.
- Disabling this feature results in **higher power consumption** since the device will not enter its lowest power state.

For optimal battery performance, we recommend keeping this setting enabled unless continuous low-power monitoring is required.

## Miscellaneous

This section covers additional topics and frequently asked questions that don't fit into other categories.

### Topics in This Section

- [Password Policy for Users](#) — Requirements and best practices for account passwords

## Other Helpful Resources

Looking for something specific? Check these related sections:

- [Common Questions](#) — General FAQ covering a wide range of topics
- [Troubleshooting](#) — Solutions for common issues
- [Support](#) — Contact information and help resources

## Password Policy for Users

### *Who this applies to*

- All customers and internal users who sign in with an email and password.

### *Sign in basics*

- Your username is your email address.
- OcuTrap uses a standard email and password sign in. There is no additional authentication at this time.

### *Password requirements*

- 8 or more characters
- Includes a number
- Includes a special character
- Do not reuse a password that you use on other sites.
- We do not block common passwords.

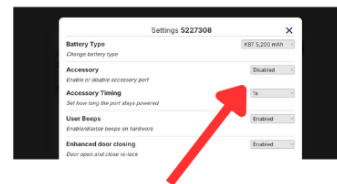
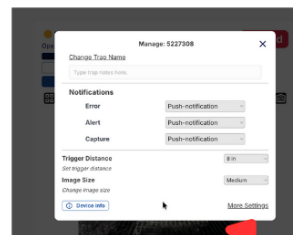
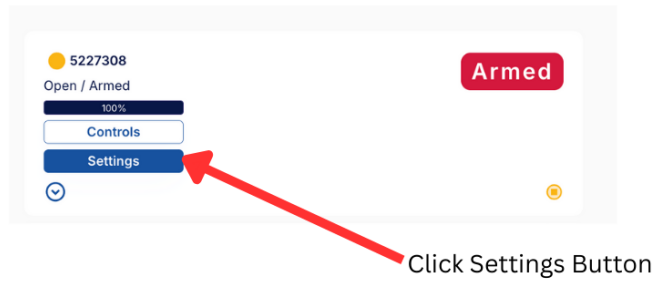
### *How to create a strong password*

- Use a passphrase with several random words, then add a capital letter and a symbol. Example: River tulip taxi folder cloud!
- Avoid personal info like names, addresses, or dates.
- Use a password manager to generate and store unique passwords.

# Accessory Port

## *Modifying Settings: Enable, Disable, and Adjust*

Learn how to manage your settings efficiently by enabling, disabling, or altering them as needed.



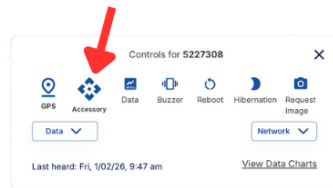
Set timing of the button

To use the button, simply click it to power the accessory.

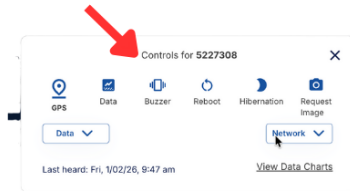


Click Controls Button

Accessory shown (enabled)



Accessory hidden (disabled)



The Accessory feature allows you to remotely power an external device connected to your OcuTrap, such as a buzzer, motor, solenoid, or other low-voltage accessory. Power is switched directly by the OcuTrap and is intended for short, controlled activations.

The accessory system consists of:

- A software-controlled **Accessory button** in the Controls menu
- A physical **12V accessory port** located on the OcuTrap pod
- Internal switching circuitry rated for up to **3.0A continuous current**

### Accessory Button Behavior

The Accessory button appears in the **Controls** panel when the accessory port is enabled.

### *When the button is visible*

- The accessory port is enabled in **Settings**
- The device is online and able to receive commands



### ***When the button disappears***

- The accessory port is disabled in **Settings**
- This is expected behavior and helps prevent accidental activation

If you do not see the Accessory button, check that **Accessory = Enabled** in the device Settings -> More Settings menu.

---

### **Physical Port Location**

The accessory port is located at the **top of the OcuTrap pod**.

- Output voltage: **12V DC**
- Maximum continuous current: **3.0A**
- Output type: Switched power controlled by OcuTrap firmware

This port is intended only for powering external accessories. It does not accept power input.

---

### **Electrical Specifications**

| Parameter              | Value                     |
|------------------------|---------------------------|
| Output Voltage         | 12V DC                    |
| Max Continuous Current | 3.0A                      |
| Switching Method       | MOSFET, low-side switched |
| Default State          | OFF                       |
| Startup Safety         | 100kΩ pull-down resistor  |

---

### **Usage Guidelines**

#### ***Polarity Sensitive***

This port is **not reversible**.

- **Pin 1:** Switched Ground
- **Pin 2:** +12V Output

Always verify polarity before connecting an accessory. Incorrect wiring may damage the connected device or the OcuTrap.

---

### ***Unidirectional Power Only***

This port is designed **exclusively to output power**.

Do **not**:

- Back-feed voltage into the port
- Connect an external power supply to the accessory pins

Back-feeding power can permanently damage the internal switching MOSFET and control logic.

---

### ***Load Requirements***

- Do not exceed **3.0A continuous current**
- High-current loads may cause overheating or failure
- Short activation times are recommended for power-hungry accessories

If your accessory requires more current, use an external relay or driver circuit.

---

### ***Inductive Loads***

If you connect any inductive device, such as:

- Motors
- Solenoids
- Relays

You **must** add external flyback protection.

- Use a diode across the load
- This protects the internal circuitry from voltage spikes during switching

Failure to add flyback protection may result in permanent damage.

---

## ***Default OFF State***

A built-in **100k $\Omega$  pull-down resistor** ensures:

- The accessory port remains OFF during boot
  - The port turns OFF if the control signal is lost
  - No accidental activation during startup or reset
- 

## **Common Use Cases**

- External buzzers
- Solenoids or latches
- Low-power motors
- Custom accessories using the OcuTrap accessory port

## ***Potential Automation Applications***

The Accessory Port enables automation and remote activation for various external devices:

1. **Rebaiter / Feeder** – Automatically dispense bait at specific intervals to keep the trap effective longer.
2. **Vaccine Feeder** – Deliver oral vaccines to target animals, aiding in disease control efforts.
3. **Lure Dispenser** – Release scent-based attractants to increase trapping efficiency for specific species.

Always test new accessories with short activations before regular use.

---

## **Important Safety Notes**

### ***Do Not Use the Accessory and Main Door Motor Simultaneously***

- The accessory port and the main door motor should not be activated at the same time because both draw power from the same 12V battery.
- The trap door motor can momentarily draw up to 5A, and if an accessory is also in use, it can exceed the power supply limits.

- **Result:** Excessive current draw may cause:
  - Device shutdown due to voltage drop
  - Overheating and potential failure of the power circuit
  - Reduced battery life

### ***Accessory Shutdown When Door Motor is Operated***

If the door motor is moved to an open or closed position while the accessory is running, the accessory will automatically power off. This behavior is an intentional safety feature designed to prevent potential conflicts between the door mechanism and the accessory. To avoid an unexpected shutdown, always ensure that the accessory is stopped before adjusting the door motor.

### ***Timing Conflicts***

- If the trap door is operating, avoid triggering the accessory at the same time.
- If using an automated system, set a delay between door operation and accessory activation in the app settings.

---

## **Troubleshooting**

### **Accessory button missing**

- Confirm the accessory port is enabled in Settings
- Refresh the page after changing settings

### **Accessory does not power on**

- Verify wiring polarity
- Confirm current draw is below 3.0A
- Check that the device is online

### **Device resets or behaves unexpectedly**

- Inspect for inductive loads without flyback protection
- Check for short circuits or overcurrent conditions

---

If you have questions about a specific accessory or need help validating your setup, contact [support@ocutrap.com](mailto:support@ocutrap.com) before deployment.

## Deleting a Trap

Clicking **Delete Trap** permanently removes your trap and all of its data. Please review the steps below before proceeding.

**Warning:** This action cannot be undone.

### Deleting a Trap

**Warning:** This action is permanent and cannot be undone. Please review the steps carefully before proceeding.

### Steps

1. **Go to Your Account Page**

Visit <https://base.ocutrap.com/account/> and scroll down to the **Devices** section.

2. **Open the Delete Trap Menu**

Click the **Delete Devices** link. A popup titled **Delete Owned Traps** will appear.

3. **Select the Trap**

Use the dropdown to select the trap you want to delete. Each entry shows the trap name followed by a unique ID.

*Only the part **before the dash** is the actual trap name—you'll need this in the next step.*

4. **Confirm the Trap Name**

Type the **exact trap name** (before the dash) into the red box to confirm. If the name does not match exactly, deletion will be cancelled.

5. **Cancel Subscription (if any)**

Any active subscription linked to the trap will be canceled automatically.

6. **Receive Confirmation Email**

You will receive an email confirming deletion, including the Trap ID for your records.

# Troubleshooting

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## Common Issues

This guide covers frequently encountered issues and their solutions based on how OcuTrap operates.

---

### Trap Won't Arm

If you're unable to arm your trap, check the following:

#### Door Must Be Fully Open

The trap **requires the door to be fully open** before arming. This is a safety feature to ensure proper capture operation.

#### Solution:

1. Open the OcuTrap app
2. Tap the **Open** button and wait for the door to fully open
3. Check that the LED shows **Solid Blue** (unarmed and open)
4. Try arming again

#### Obstruction Detected

Before arming, the trap performs an **obstruction check** to ensure the capture zone is clear. If something is blocking the sensor, arming will fail.

#### Solution:

1. Check that nothing is in front of the sensor inside the trap
2. Clear any debris, leaves, or objects from the trap interior
3. Ensure the sensor window is clean
4. Wait for 5+ distance readings to confirm the zone is clear
5. Try arming again

## Motor Connectivity Issue

The trap tests motor connectivity before arming. If the motor doesn't respond, arming will fail.

### Solution:

1. Check the motor connector is securely attached
  2. See [Motor Connector Tightness Check](#)
  3. Verify the motor cable isn't damaged
  4. Contact support if the issue persists
- 

## False Triggers / Unwanted Captures

OcuTrap has sophisticated false-trigger prevention, but environmental factors can sometimes cause issues.

### Rain or Debris Triggering

Heavy rain or debris falling through the trap can sometimes trigger captures.

### How OcuTrap Prevents This:

- Requires **3+ consecutive valid readings** before triggering
- **Oscillation detection** identifies rain patterns (rapidly changing distances)
- **Signal quality filtering** rejects weak or noisy readings

### If you're still getting false triggers:

1. Increase the **Capture Distance** setting (move trigger point further from sensor)
2. Ensure the trap is positioned to minimize rain entry
3. Check that the sensor window is clean and undamaged
4. Consider repositioning the trap to a more sheltered location

### Capture Distance Too Sensitive

If the trap triggers before animals are fully inside:

### Solution:

1. Go to **Settings** → **More Settings**

2. Decrease the **Capture Distance** value (smaller = animal must be closer)
  3. Default is 250mm (10 inches) — try 200mm or 150mm for more selective triggering
- 

## GPS Not Updating

GPS updates are battery-optimized and may not update as frequently as expected.

### Understanding GPS Behavior

- **Default interval:** Every 8 hours (not real-time)
- **First boot delay:** 5-minute delay before first GPS acquisition
- **Capture updates:** GPS automatically updates when a capture occurs

### GPS Shows Old Location

#### Solution:

1. Wait for the next scheduled update (check your GPS Interval setting)
2. Request a manual update: Go to Controls → tap the Data button
3. Ensure GPS is not disabled in settings

### No GPS Fix Available

#### Solution:

1. Ensure the trap is outdoors with a clear view of the sky
  2. Move away from buildings, dense tree cover, or metal structures
  3. Allow up to 3 minutes for the first fix after power-on
  4. Check that GPS is enabled in settings
  5. If problems persist, try a factory reset of the GPS module (contact support)
- 

## Camera Issues

### Dark or Black Images

#### Possible Causes:

- Camera not detecting darkness correctly



- IR LEDs not activating

### Solution:

1. Check **Dark Lux Threshold** setting — lower value = IR activates at higher light levels
2. Increase **Minimum IR Brightness** setting
3. Ensure the IR LED window is clean
4. Verify the camera lens is not blocked or dirty

### Overexposed / Washed Out Images

### Solution:

1. Decrease **Maximum IR Brightness** setting
2. Adjust image cropping to remove reflective areas
3. Reposition the trap to reduce direct reflections

### Images Not Sending

### Possible Causes:

- Poor cellular signal
- Large image size taking too long to transfer

### Solution:

1. Check cellular connectivity (LED should be breathing cyan when connected)
  2. Reduce **Camera Quality** setting (1-2 for faster transfer)
  3. Move trap to an area with better cellular coverage
  4. Wait — images transfer in 8KB chunks and may take time on slow connections
- 

## Connectivity Issues

### Trap Shows "Offline"

**OcuTrap's Auto-Recovery:** The trap has automatic stuck-offline detection. After 20 minutes offline, it will:

1. Disconnect from the network

2. Power cycle the cellular modem
3. Attempt to reconnect
4. Retry with increasing intervals (10, 20, 30... up to 60 minutes)
5. Reset the system after 5 failed attempts

**If the trap stays offline:**

1. Check battery level — low battery can affect connectivity
2. Verify cellular coverage in the deployment area
3. Press the power button to wake the device
4. Contact support if the trap remains offline for extended periods

**Commands Not Reaching Trap**

See [Trap Not Sending Commands](#) for detailed troubleshooting.

---

**Battery Issues****Battery Draining Quickly****Common Causes:**

- Poor cellular signal (device uses more power searching)
- GPS interval set too frequently
- Camera timelapse interval set too short
- Cold temperatures reduce battery capacity

**Solution:**

1. Deploy in areas with good cellular coverage
2. Increase GPS Interval (8+ hours recommended)
3. Increase Camera Time Lapse interval (6+ hours recommended)
4. In cold weather, expect reduced battery life
5. Keep firmware updated (includes battery optimizations)

**Trap Keeps Hibernating**

If the trap enters hibernation unexpectedly:

1. **Check battery voltage** — hibernation occurs below 9.6V (default)
2. **Charge or replace the battery**

3. **Verify the correct Battery Type** is selected in settings
  4. If the battery is charged but hibernation persists, the battery may be damaged
- 

## **Door Issues**

### **Door Won't Open or Close**

#### **Solution:**

1. Check motor connector is securely attached
2. Verify no physical obstruction is blocking the door
3. Check battery level — door operation requires adequate power
4. Use the manual door control: Double-press User Button + hold for 5 seconds
5. Check for motor fault indicator (orange LED)

### **Door Opens/Closes Slowly**

#### **Possible Causes:**

- Low battery
- Motor wear
- Mechanical obstruction

#### **Solution:**

1. Charge the battery fully
  2. Check for debris in the door track
  3. Contact support if the issue persists
- 

## **LED Indicators**

### **No LED / Trap Appears Dead**

#### **Solution:**

1. Press the power button to wake from hibernation
2. Charge the battery — the trap may have auto-hibernated due to low power
3. Check battery connections

4. If battery is charged and power button doesn't respond, contact support

### **Rapid Red Blinking (SOS)**

This indicates a firmware crash.

#### **Solution:**

1. If fewer than 10 blinks, the trap may recover automatically
2. If more than 10 blinks, contact support
3. Note any patterns or counts to share with support

For LED status meanings, see the [LED Guide](#).

---

### **Tilt / Movement Alerts**

#### **Unexpected Tilt Alerts**

The trap sends tilt alerts when not level during armed mode.

#### **Solution:**

1. Ensure the trap is placed on level ground
  2. Check that the trap is stable and not shifting
  3. Only one tilt alert is sent per arming session to prevent spam
- 

### **Still Need Help?**

If you've tried the solutions above and still have issues:

1. [Contact Support](#) with details about your issue
2. Include: trap serial number, battery voltage, LED status, and steps already tried

### **Trap Not Sending Commands**

Issue

If your OcuTrap is not responding to commands and is displaying error codes, it may have entered hibernation mode or lost connection.

### **Possible Causes**

1. **Hibernation Mode** – The trap automatically enters hibernation when it is not armed or in a captured state. If it was set but not fully armed, it will eventually power down to conserve battery.
2. **No Battery Connection** – Ensure the battery is properly connected and charged. A loose or depleted battery will prevent the trap from operating.
3. **No Trap Online** – If the trap is not showing as online, it may not be connected to the network.
4. **Poor Cellular Communication** – Weak signal strength can prevent the trap from sending or receiving commands.

### **Solutions**

#### ***Wake the Trap from Hibernation***

- Press the **power button** (the left-most button) to wake up the trap.
- Ensure the trap is in **armed mode** to prevent it from entering hibernation again.

#### ***Check Battery Connection***

- Verify that the battery is properly seated and securely connected.
- Check the battery charge level and recharge or replace if necessary.

#### ***Verify Network Connectivity***

- Check if the trap appears online in the app or dashboard.
- Move the trap to an area with better cellular reception if the signal is weak.

### **Additional Help**

If the issue persists after trying these steps, please contact support for further assistance.

## Troubleshooting: Motor-to-Pin Connection Too Tight

### Issue

When attaching the motor to the bracket:

- The **pin and clevis** may not fit easily into the motor connector.
- This can cause frustration during setup and may appear as though the parts are misaligned or stuck.

### Example scenario:



### Cause

This issue is usually due to the **motor being fully retracted** at the time of assembly. In its fully retracted state, the motor's position prevents easy insertion of the clevis and pin.

**Solution**

To solve this:

1. **Power on the OcuTrap.**
2. Using the **app** or the **buttons inside the POD, press the "Close" button** to extend the motor slightly.
  - This will expand the motor outward, creating enough clearance for the pin and clevis to fit properly.
3. **Prop up the motor** to allow it room to extend fully (see images below).

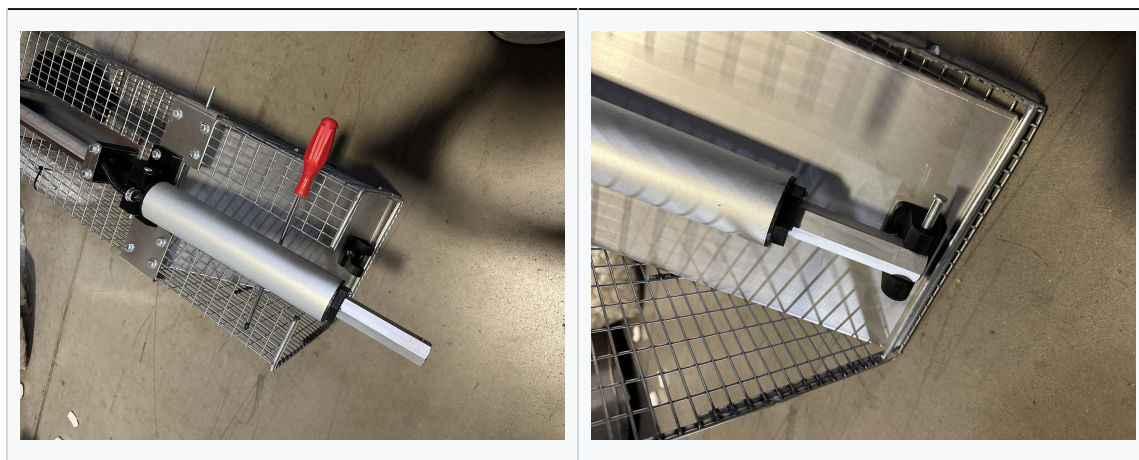




## Step-by-Step

| Step | Action                                                                                     |
|------|--------------------------------------------------------------------------------------------|
| 1    | Ensure the OcuTrap is powered on and motor is plugged in.                                  |
| 2    | Use the app or POD controls to press the <b>Close</b> button, extending the motor outward. |
| 3    | Hold or prop up the motor (lift it slightly) to give it clearance to move.                 |
| 4    | Once extended, <b>let the door down all the way</b> and slide it forward to reposition.    |
| 5    | Unplug the motor to prevent it from moving unexpectedly.                                   |
| 6    | Insert the clevis and pin into the motor connector easily.                                 |

### Photos for reference:



### Tip

- Always make sure the **trap door is fully lowered** before attempting final assembly of the motor to the bracket.
- If the motor is already fully extended, no further action is needed.

## Connection Issues

If your OcuTrap is having trouble connecting, follow these steps to troubleshoot the issue:

### 1. Check the Battery

- **Ensure the battery is fully charged** and properly connected.
- If the battery is not fully charged, charge it and reconnect it to the device.
- Instructions on how to charge:

- Plug the two yellow connectors together to supply

## ***2. Inspect the LED Light***

- If **there is no LED light** on the trap, verify the wiring and connections.
- Make sure all wires and connectors are **securely attached** and fully seated.

## ***3. Verify Configuration in Your Account***

- Log in to your OcuTrap account and confirm that the trap is listed in your **Main Console**.
- If the trap does not appear, go to the **Account Page** and **add the trap** to your account.

## ***4. Match the Serial Number (SN)***

- If the trap is listed in the console, verify that the **Serial Number (SN)** on the trap matches the **Particle chipset SN** visible in the trap's settings.
- To do this, click the **Settings button** at the bottom left of the screen and check the SN against the one displayed on the Particle chipset (see image).

## ***5. Check Cellular Range***

- If the **LED light is blinking green**, the trap may be out of cellular range.
- Ensure you are in an area with **sufficient cellular coverage**.

## ***6. Inspect Antenna Connections***

- If the issue persists, check that the **gold antenna connections** (as shown in the image) are fully pressed down and securely attached to the trap.

## Condensation on the Camera

### 1. How to Spot the Issue

| Symptom                                                       | Likely Cause                                                                  | Severity |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|----------|
| <b>Foggy or “milky” images</b> that clear after a few minutes | <b>External condensation</b> —morning/evening temperature swings              | Low      |
| Persistent haze or water beads <b>inside</b> the lens window  | <b>Internal moisture</b> —air leak at the lens gasket or loose accessory port | Medium   |
| Visible droplets <b>inside</b> the device                     | <b>Water ingress</b> —damaged seal or submerged trap                          | High     |

### 2. Immediate Fix (10 min)

#### 1. Power down & open the POD

- Tap **Settings** ► **Power Off** or hold the left-most button until the LEDs go dark.
- Remove the battery pack and disconnect the accessory cable (if attached).

#### 2. Wipe the outside glass

Use a microfiber cloth or lens wipe. If fogging disappears, the issue was external.

#### 3. Inspect the lens seal

- Look for debris, hair, or grit around the rubber gasket.
- Gently clean the seal with isopropyl alcohol and a lint-free swab.

#### 4. Dry packs save the day

Place a fresh silica gel pack inside the trap pod before closing.

### 3. Deep-Dry Procedure (24–48 hrs)

If internal moisture persists:

1. Remove the **battery**
2. Leave the POD cracked open in a warm, ventilated spot for **24 hrs** —or—  
Seal the POD in a zipper bag with **desiccant** packs for **12–18 hrs**.
3. Re-assemble, close the POD, and request an image to confirm clarity.

#### 4. Prevention Checklist

- **Gasket care** – Lightly coat the rubber seal with silicone grease every 6 months.
- **Tight hardware** – Finger-tighten the antenna and accessory caps after each service visit.
- **Avoid submersion** – OcuTrap R1 is weather-sealed, but not rated for full underwater exposure.

#### 5. Still Seeing Haze? We're Here to Help.

1. Take **two clear photos** of the issue: one of the lens/window and one of the interior POD.
2. Email [support@ocutrap.com](mailto:support@ocutrap.com) with the photos, trap id, and a brief description.
3. Our technicians will review within **1 business day** and advise the next step:

| Scenario                                       | Cost to You                                             |
|------------------------------------------------|---------------------------------------------------------|
| <b>Under warranty</b> & seal failure confirmed | Free repair or replacement + prepaid return label       |
| Out of warranty, but repairable                | Flat service fee <b>\$25</b> (includes return shipping) |
| Severe water damage / corrosion                | Discounted upgrade to the latest revision               |

#### 6. Warranty Quick Facts

- Standard coverage: **12 months** from delivery.
- Exclusions: submersion, impact damage, or housing modifications.

### Motor Connector Use

► **This page has a video.** Watch it online at [docs.ocutrap.com](https://docs.ocutrap.com).

## Motor Connector Tightness Check

### *Symptoms*

- Motor does not move at all
- Door does not open or close when commanded
- Motor can be heard engaging but produces no motion
- Intermittent operation after handling or transport

### *Cause*

The motor is **not tightened enough**.

If the motor mounting hardware is too loose, the motor shaft does not stay properly aligned under load. This prevents effective torque transfer and can cause the motor to slip or fail to engage.



### *Correct Gap Specification*

The motor must be tightened to leave a small, controlled gap.

- **Required gap:** approximately **0.15 to 0.20 inches**
- Less than this can cause motor electronics not to connect properly

- More than this usually means the motor is too loose

A gap larger than 0.20 inches is a common cause of the motor not operating.

### ***How to Check***

1. Inspect the motor mounting location.
2. Verify the motor is not loose or able to shift by hand.
3. Measure the gap between the motor and the mounting surface.
4. Confirm the gap is within the 0.15 to 0.20 inch range.

### ***How to Fix***

1. Tighten the motor mounting hardware gradually.
2. Stop tightening once the gap is within 0.15 to 0.20 inches.
3. Ensure the motor is firmly secured and cannot move under hand pressure.
4. Do not overtighten past this range.

### ***After Adjustment***

After tightening the motor and setting the correct gap, try controlling the door once again

If the motor still does not operate after this adjustment, disconnect and reconnect the motor connector and repeat the steps.





# Support

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## Contact Us

### Chat Support

You can message OcuTrap support through the chat bubble that can be found on the [OcuTrap.com](https://ocutrap.com) homepage by clicking the chat icon on the bottom right of the page.

### Email Support

You can send us an email through the [contact page](#) or emailing us at [info@ocutrap.com](mailto:info@ocutrap.com)

### Website Status

Check if our website is down at [ocutrap.statuspage.io](https://ocutrap.statuspage.io)

## Bug reporting

If you find a bug, please report it quickly at our portal: [base.ocutrap.com/bug\\_report](https://base.ocutrap.com/bug_report). Include a detailed bug description, reproduction steps, and any relevant visuals. Explain what you expected to happen and what actually did. Your detailed feedback helps improve our service.

You can also email [info@ocutrap.com](mailto:info@ocutrap.com)



## Safety Information

Using the OcuTrap involves inherent risks. Always adhere to the guidelines in this manual to ensure safe operation and prevent injury or equipment malfunction.

### Important:

This manual should be read in its entirety prior to using the OcuTrap, in order to prevent any serious injury or harm. Failure to review this Operations Manual may result in injury.

---

### Warning: Hazard

- **Hazard:**

This device uses a depth sensor to remotely trigger the actuator on the door. This mechanism allows the door to close quickly, which can cause serious injury if precautions are not followed.

**Ensure that nothing is blocking the door's path**, as this may not only cause injury but can also lead to trap malfunction.

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### Child Safety

- **Child Safety Warning:**

This product can cause serious injury and contains small components. **Keep children away from the equipment** at all times.

---

### Operational Notice

- **Power Management:**

Make sure that your OcuTrap is turned off when not in use to preserve battery power.

- **POD Security:**

Keep the OcuTrap POD closed after setup to prevent any malfunctions in the system.

- **Water-Tight Assurance:**

Verify that the POD is water-tight during operation by closing both latches and tightening the knob on the right side of the device.

---

## **Potential Injury: Finger and Hand Damage**

### ***Overview***

The OcuTrap door is designed to open and close automatically and swiftly. While this is essential for its proper operation, it also poses a risk of injury—especially to fingers and hands—if safety precautions are not strictly observed.

### ***Potential Injuries***

- **Finger Injuries:**

The rapid closure of the door can result in pinching, crushing, or lacerations.

- **Hand Injuries:**

Similarly, if a hand is in the door's path, it can suffer comparable injuries.

### ***Safety Guidelines***

- **Keep a Safe Distance:**

Always ensure that your fingers, hands, and any other body parts are completely clear of the door's path before the trap is activated.

- **Supervise Operation:**

Keep children and pets away from the device during its operation to prevent accidental injury.

- **Review the Manual:**

Familiarize yourself with all aspects of this manual and the device's operation to fully understand the safety features and risks.

- **Regular Maintenance:**

Inspect the device frequently to ensure that the door mechanism and other components are functioning properly without signs of wear or misalignment.

- **Emergency Stop:**

Learn and utilize the emergency stop feature (if available) to quickly disable the door mechanism in the event of an emergency.

---

## In Case of Injury

- **Immediate Action:**

Stop using the device immediately. Administer basic first aid such as cleaning the wound and applying a bandage.

- **Seek Medical Attention:**

For serious injuries, persistent pain, or significant swelling, contact a medical professional immediately.

- **Report the Incident:**

For additional support or to report an incident, please contact our support team at [support@ocutrap.com](mailto:support@ocutrap.com).

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### Note:

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## Purchases

### Payments and Purchases

Payments and purchases can only be done through the website at [www.ocutrap.com](http://www.ocutrap.com) on your mobile phone or computer.

## Nonprofit and 501(c) Program

### Nonprofit and 501(c) Program

OcuTrap partners with wildlife rescues, shelters, conservation groups, municipalities, and schools. Eligible 501(c) organizations can receive discounted pricing and tax exempt purchasing where allowed by state law.

**Apply here:** <https://ocutrap.com/pages/contact>

---

### Benefits

- **Discounted hardware pricing.** Special nonprofit rates on traps and accessories.
- **Onboarding and training.** Setup help, volunteer training materials, and safety guides.

- **Priority support.** Faster responses and a dedicated point of contact for larger deployments.
- **Flexible purchasing.** Quotes on request. Purchase orders and invoicing for qualifying organizations.
- **Bulk benefits.** Volume pricing and spare parts discounts for multi-trap fleets.

#### **Need something not listed**

Tell us what would help your mission. We can tailor training, reporting, and deployment plans.

---

### **Eligibility**

- U.S. 501(c) organizations such as 501(c)(3) charities and 501(c)(4) social welfare groups.
- Government agencies and accredited schools may qualify for nonprofit pricing.
- Sales tax exemption depends on the shipping destination state. A state sales tax exemption certificate is usually required.

---

### **How to get the nonprofit discount**

1. **Submit the application.** Use the contact form and title form “Nonprofit pricing and tax exemption.”  
Apply at <https://ocutrap.com/pages/contact> or email [support@ocutrap.com](mailto:support@ocutrap.com)
2. **Share basic info.** Organization legal name, EIN, billing and shipping addresses, website, and a primary contact.
3. **Account review.** We verify your details and flag your customer account for nonprofit pricing.
4. **Sign in to order.** Your nonprofit pricing appears on quotes or invoices when you shop or place POs.

**Processing may take 1-2 weeks**

---

### **Steps to receive the tax exempt rate**

1. **Collect state documentation.**  
Provide a **state sales tax exemption certificate** for each state where you want tax exempt purchasing.

2. **Send certificates for approval.**

Upload during the application or include links when you contact us.

3. **Verification.**

We review and mark your account as **tax exempt** for the covered states.

4. **Place orders while signed in.**

Sales tax will not be charged on qualifying orders shipped to approved states.

5. **Keep certificates current.**

Renew before expiration to avoid tax on future orders.

***Required documents checklist***

☐

IRS determination letter showing your 501(c) status

☐

State sales tax exemption certificate for each shipping state

☐

EIN and organization legal name

☐

Billing and shipping addresses

☐

Website URL and primary contact details

**Note**

The IRS 501(c) letter confirms nonprofit status for discounts. It is not enough for state sales tax exemption in most states.

---

**Purchasing options**

- **Quotes and POs.** Available for qualifying organizations.
  - **Invoicing terms.** Net terms may be available after review.
  - **Credit card or ACH.** Supported for faster fulfillment.
- 

**FAQs**

**Is the IRS 501(c) letter enough for tax exempt purchases**

Not in most states. You will need the applicable state sales tax exemption certificate.

**Do municipalities and schools qualify**

Yes. State tax exemption depends on your state and documentation.

**Can we use a purchase order**

Yes, for qualifying organizations. Include PO details when you contact us.

---

**Contact**

Questions or ready to apply

**<https://ocutrap.com/pages/contact>**

# Legal & Compliance

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## Warranty Information

<https://ocutrap.com/pages/warranty>

## Legal disclaimers and compliance information

### Terms of Service

<https://ocutrap.com/policies/terms-of-service>

### Privacy Policy

<https://ocutrap.com/pages/privacy-policy>

### Hardware Warranty

<https://ocutrap.com/pages/warranty>

### Software License

<https://ocutrap.com/pages/software-license-usage-terms>

### Refund Policy

<https://ocutrap.com/policies/refund-policy>

### Animal Recognition Policy

<https://ocutrap.com/pages/ocutrap-image-recognition>

# Device Management

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## Selling or Transferring a Trap

### Step 1: Remove the Trap From Your Account

1. Log in to your account at [base.ocutrap.com/account](https://base.ocutrap.com/account).
2. Locate the trap you wish to transfer.
3. Click the **Delete Trap** button.
4. Confirm the trap name exactly as prompted.
5. The trap will be unlinked from your account.

For full instructions, including what happens to your subscription and trap data, see

### Step 2: New Owner Adds the Trap

1. The new owner must have their own OcuTrap account.
2. After logging in, they should go to the **Account Page** and scroll to the bottom.
3. Click the **Add Trap** button.
4. Enter the **Trap ID** (serial number), found inside the trap's POD cover.

### Step 3: Subscription and Setup

- The new owner will be prompted to activate a subscription plan if needed.
- They can now customize settings, view live data, and receive notifications from the trap.

---

### Important Notes

- The **warranty** begins from the date the trap was first activated, not the transfer date.
- Any **free trial** of service may have already been used during the original activation.



## Trap Tests

**Firmware requirement:** v565 or later

A safe, audio-only demo of your trap's detection pipeline — no door movement.

---

### Before You Start

Confirm all four before running Trap Test:

1. Trap door is **open**
2. Trap is **unarmed**
3. Trap is **clear** of obstructions
4. Open the back pod door so the beeps can be clearly heard

### Finding Trap Test Button

From your trap, tap **Settings** → **More Settings** → **Trap Test** button at the top of the settings page.

### What the Button Does

Tapping **Trap Test** sends a test command to your device. The trap enters a safe demo mode — audio feedback only, no door movement. There is no visible response in the app; listen for audio from the trap to confirm the test is running.

### Running the Test

Wave your hand (or any object) toward the sensor. You'll hear the trap react through four phases:

| Phase            | What You Hear       | What It Means                                                |
|------------------|---------------------|--------------------------------------------------------------|
| <b>Waiting</b>   | Silent              | Ready — move something toward the sensor                     |
| <b>Detecting</b> | Beeps, speeding up  | Object seen within ~39 in (1000 mm)                          |
| <b>Verified</b>  | Confirmation cue    | Reading is stable — move into the capture zone               |
| <b>Capture!</b>  | Solid 3-second tone | Crossed the trigger distance — door would close in real mode |

# Account and Billing

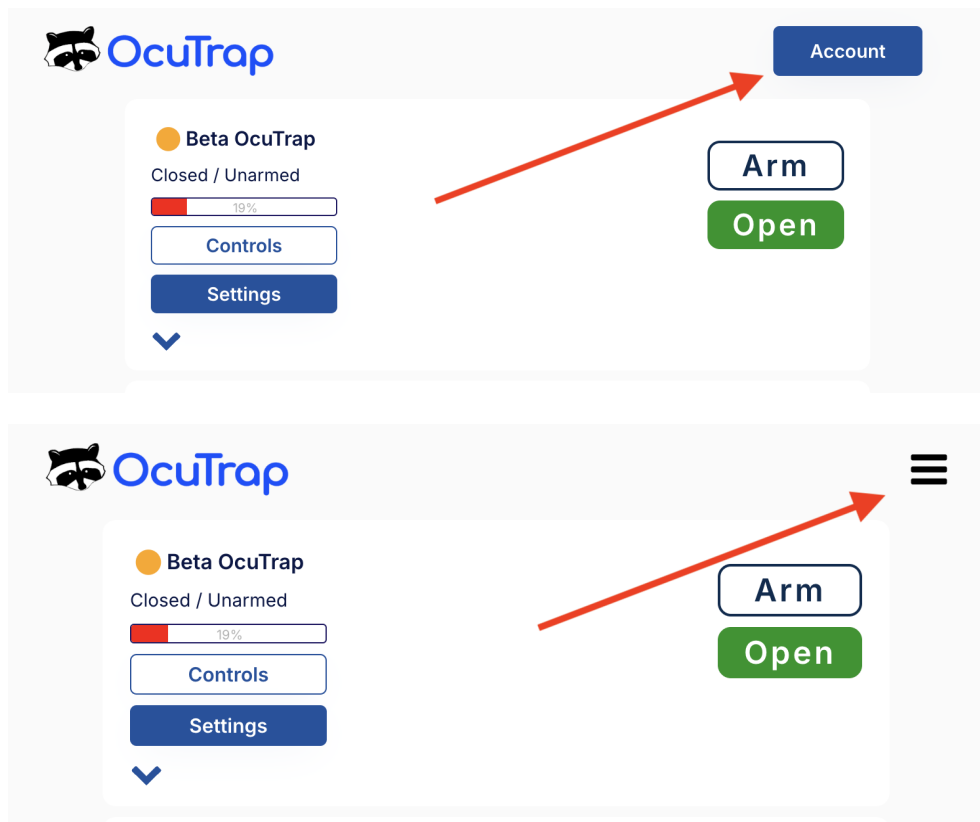
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## Billing

### 1. Accessing Your Account Page

1. **Log in** to your OcuTrap account at [base.ocutrap.com](https://base.ocutrap.com).
2. Once logged in, either:
  - Click the **Account** link in the navigation bar, **OR**
  - Click the **burger menu** (three horizontal lines) and select **Account**.

You will be taken to your **Account** page, which displays personal information (name, email, phone number), security settings, and billing information.



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### 2. Navigating to the Billing Portal

1. From the **Account** page, scroll to the **Billing** section.

2. Click on **Manage Subscriptions** to open the secure billing portal.



This portal allows you to view and manage all aspects of your subscription, including:

- **Updating Billing Information** (address, payment method, etc.)
  - **Managing Credit Cards** (add/remove payment methods)
  - **Changing Your Plan** (Monthly, Annual)
  - **Viewing Subscription Status** (active, trialing, canceled, etc.)
- 

### 3. Managing Your Subscriptions

Once you are in the secure billing portal, you'll see an overview of your subscriptions for your traps. From here, you can:

#### 3.1 Update Billing Information

- **Address** – Update your physical billing or shipping address.
- **Credit Cards** – Add, remove, or modify your credit card on file.
- **Payment Methods** – Choose a default payment method if you have multiple.

#### 3.2 Change Subscription Plans

- **Upgrade** – Move to a higher-tier plan that includes more features or traps.
- **Downgrade** – Switch to a lower-tier plan if you need fewer features or traps.
- **Cancel** – End your subscription. You'll still have access for the remainder of your billing cycle unless otherwise noted.

#### 3.3 Plan Status Definitions

- **Active**

Your subscription is currently active, and you have full access to all features.

- **Trialing**

Your subscription is in a free trial period. You won't be charged until the trial ends (unless you decide to cancel or change plans before then).

- **Canceled**

Your subscription has been canceled. You'll retain access until the cancellation date takes effect.

Depending on your plan or provider settings, you might see other statuses such as **Past Due**, **On Hold**, or **Paused**. If you have questions about these statuses, [contact our support team](#).

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## 4. Frequently Asked Questions

### *4.1 How do I update my payment method?*

1. Go to **Account** → **Manage Subscriptions**.
2. Select **Billing** or **Payment Methods**.
3. Add a new card or edit existing card details.
4. Confirm your changes.

### *4.2 Can I view my billing history?*

Yes. In the billing portal, you'll typically find a **Billing History** or **Invoices** tab that provides a record of all past charges.

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## 5. Getting Help

If you encounter any issues or have additional questions, please reach out to us:

- **Email:** [support@ocutrap.com](mailto:support@ocutrap.com)

We're here to help you get the most out of your OcuTrap subscription.

## Changing Your Subscription Payment Method

### *1. Log in to Your Account*

1. Visit **OcuTrap.com** and sign in to your account.

2. Go to **Account Page**
3. Scroll to **Manage Subscription** page.
4. This will take you to the **Stripe customer portal** where your subscription details are stored.

## ***2. Add a New Payment Method***

1. In the Stripe portal, scroll down to **Payment Methods**.
2. Click **Add payment method**.

## ***3. If You Are Using Link***

Some customers already have a card saved through **Link**, Stripe's fast checkout system.

If you see that your card is associated with Link:

1. Click the **three dots** at the top right of the payment method.
2. Select **Pay without Link**.  
This will allow you to use a normal card or bank account instead of Link.

## ***4. Enter Your New Payment Information***

You will now see options to:

- Add a new credit or debit card
- Add a bank account Enter your details and save the payment method.

## ***5. Set the New Method as Default (Optional)***

If you entered multiple payment methods, choose which one should be used for your subscription.

1. Under the payment method list, click **Make default** next to the method you want active.

## Update Individual Trap Subscriptions

### When You Might Need to Update

- The **Plan** shows **Canceled** or is **blank** in your trap's Device Info
  - You've never started a subscription for this trap
  - Your subscription ended or payment failed
- 

### Update Your Subscription in a Few Steps

1. **Log in to the OcuTrap Console**

Go to [your console](#) and sign in.

2. **Open Your Trap Settings**

Find the trap in your list and click **Settings**.

3. **Check the Plan Status**

In the Device Info section, look for the **Plan** field.

- If it says **Canceled** or **No Subscription**, you'll see an **Update** link next to it.

4. **Click Update**

Click **Update** to start the subscription process for this trap.

5. **Complete Checkout**

Choose your plan, enter payment details, and confirm.

Once finished, your plan will show as **Active**, and your trap will continue to work without interruption.

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**Tip:** Each trap has its own subscription, so if you have more than one, repeat these steps for each trap you want to activate. You can see all your active subscriptions in your account's Stripe portal link.

## Resetting Password

To reset your password, follow these concise steps:

1. Visit the [Password Reset Page](#).

2. Select "forgot password"

- The **forgot password** option will guide you through the standard reset process.

- The **magic link** provides a secure, one-time login, sent to your email, expiring after one hour for enhanced safety.

## Managing Your Subscription

### How to Access Subscription Settings

1. Go to [base.ocutrap.com](https://base.ocutrap.com)
2. Log in with your account credentials.
3. Click on **Account** in the menu.
4. Select **Manage Subscription**.
5. Choose the trap you want to update.

### Available Options

Within the subscription portal, you can:

- **Switch plans** – Change from **Monthly** to **Annual** billing (and vice versa).
- **Update payment method** – Add a new card or edit your existing payment details.
- **Cancel subscription** – Stop your subscription at any time. You'll continue to have access until the end of your current billing period.

### Notes

- All billing is processed securely through Stripe.
- If you cancel, you won't be charged again, but your subscription remains active until the billing period ends.
- Annual subscriptions are billed once per year, offering savings compared to monthly billing.

## Account Deletion

### *Account Deletion Guide*

To permanently delete your account, please follow the steps below:

1. Visit the [Account Deletion Page](#).
2. Click on the **Delete Account** button. If prompted, you may need to log in.

3. Type **Delete** in the input box that appears. Remember, it's case sensitive.
4. Click on the **Delete Account** button again to confirm your action.

Your account will then be fully deleted.